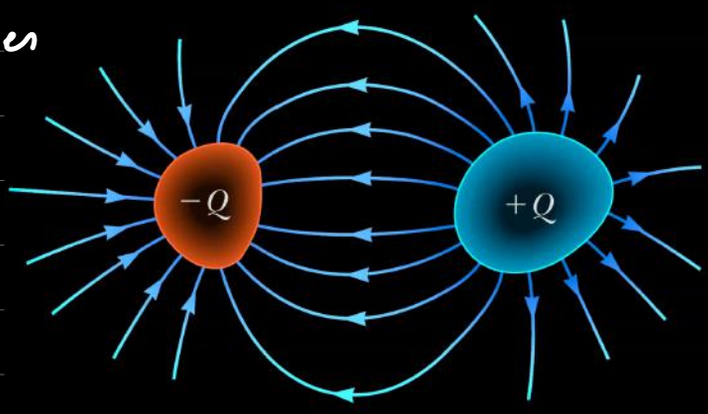
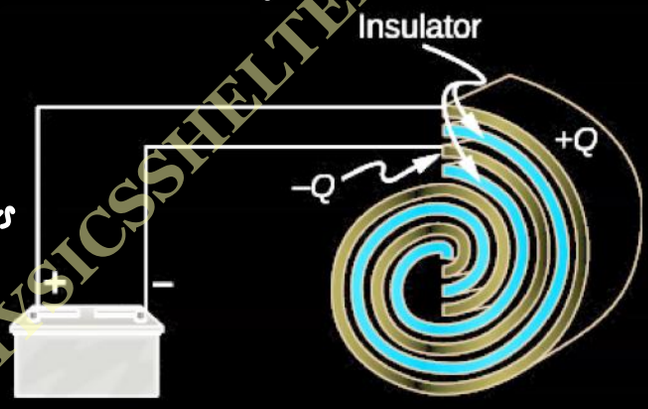
CAPACITOR: Two arbitrary conductors kept at some separation from each other form a capacitor. Typically the two conductors are given charges of equal magnitude and opposite polarities, however in some problems we come across unequal magnitude charges as well. Sometimes we also visualise a capacitor made of a single conductor also, in such a case we imagine the other conductor to be at infinity (imagine a conducting shell of infinite radius. Sometimes the two conductors may be separated by insulators (or dielectrics) other than free space (for example adjacent figure shows capacitor made of two rolled conductors separated by a rolled insulating layer).





CAPACITANCE: When two conductors forming a capacitor are given charges $+Q$ and $-Q$ and the potential difference between them is V, then we define capacitance of the system of conductors as:

$$\boxed{C = Q/V}$$ Units: Farad

→ Even though $C = Q/V$, it neither depends on Q nor on V, rather it is purely a geometrical factor. (Consider the charges on both conductor to be increased to η times then by uniqueness, the distribution is still similar to the original distribution. This will scale up the field everywhere in space by same factor, and hence potential difference will also become ηV. Thus $C = \eta Q/\eta V$ and it is still Q/V.

→ By charge of a capacitor we mean magnitude of charge on one of the conductor (net charge is zero in most situations).

YOUTUBE.COM/@THEPHYSICSSHELTER

Calculating Capacitance :

Step I: Give $+Q$ & $-Q$ charges on conductors

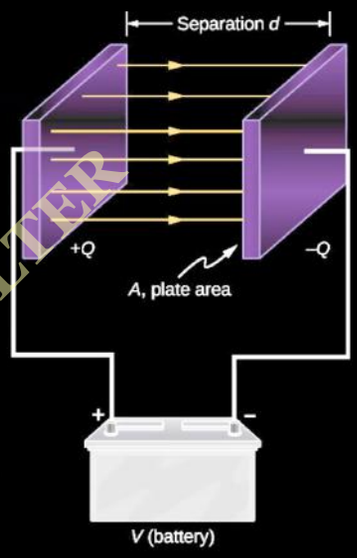
Step II: Find field in the region between conductors

Step III: Find $\int \vec{E} \cdot d\vec{s}$ from positive conductor to the negative conductor. This is V. Now $C = Q/V$.

Parallel Plate Capacitor:

For most purposes we model the field between parallel plates as the one generated by infinite parallel plates. Now field between two plates:

$$\vec{E} = \frac{Q}{2A\epsilon_0} \times 2\hat{i} = \frac{Q}{A\epsilon_0}\hat{i}$$



Now work done in taking unit charge from +ve plate to negative plate, $V = Ed = Qd/A\epsilon_0$

Thus $C = Q/V = \epsilon_0 A/d$. Thus for || plate Capacitor:

$$\boxed{C = \epsilon_0 A/d}$$

Force Of Attraction Between Parallel Plates:

Force is equal to field due to one conductor multiplied by the charge of the other conductor. Thus:

$$F = \frac{Q}{2A\epsilon_0} \cdot Q \Rightarrow \boxed{F = Q^2/2A\epsilon_0}$$

Energy Stored In A Capacitor: The electrostatic energy of a system is negative of work done by electric field (or work done by external agent in slowly arranging the system without change in KE). Now let C be the capacitance of a system. Let the conductors have instantaneous charges $+q$ and $-q$, then instantaneous potential difference, $V = q/C$. Now suppose we move an additional dq from negative conductor to the positive conductor then:

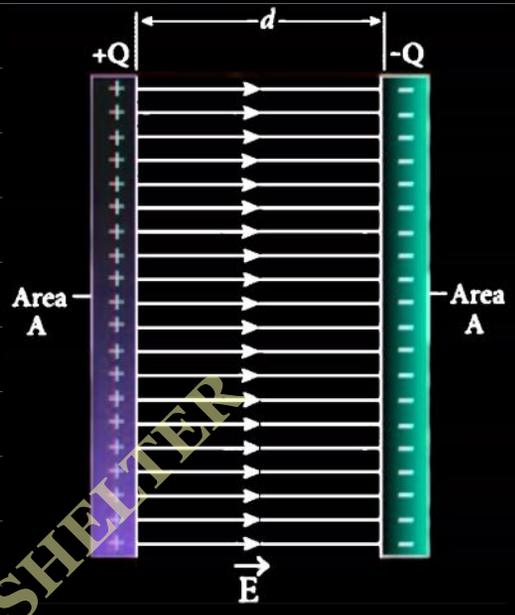
$$dU = V\,dq = q\,dq/C,$$ now suppose we wish to shift a total charge Q from one initially neutral conductor to the other conductor (also initially neutral). Then total potential energy accumulated in the process:

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

$$U = \int dU = \int_0^Q q\,dq/C = Q^2/2C, \quad \text{thus for any capacitor:}$$

$$\boxed{U = Q^2/2C = \tfrac{1}{2}QV = \tfrac{1}{2}CV^2} \quad (\because C = Q/V)$$

Energy Density In A Parallel Plate Capacitor:



Volume $= \forall = Ad \Rightarrow$ Energy Density:

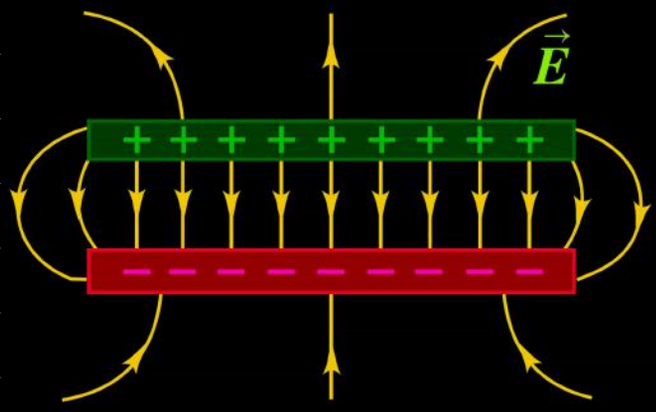
$$u = U/\forall = \tfrac{1}{2}CV^2/\forall$$

$$\Rightarrow u = \frac{1}{2}\frac{\epsilon_0 A}{d}(Ed)^2 \cdot \frac{1}{Ad} = \frac{1}{2}\epsilon_0 E^2$$

Thus $\boxed{u = \frac{1}{2}\epsilon_0 E^2}$ $(\because V = Ed)$

We have proved the result for electrostatic Energy Density in a parallel plate Capacitor, however same result holds true for energy density anywhere in free space irrespective of how that electric field is generated.

Fringing of field lines in || plate capacitor



While for simplicity we had modelled the field due to parallel plate capacitor like a system of infinite plates, actually near the end of plates there is a distortion of field lines, which becomes an important consideration in some analyses (such as suction forces on dielectrics, which will be discussed later). For a point at a very large distance from the capacitor, the capacitor will behave like a point dipole.

YOUTUBE.COM/@THEPHYSICSSHELTER

Concentric Cylindrical Shell Capacitor:

We imagine the capacitor to be sufficientlly long so that field may be taken as radial. Now by shell theorem (for cylindrical shells),

$$E = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{Q}{2\pi l \epsilon_0 r}$$

$$\Rightarrow V = \int_a^b E\,dr = \frac{Q}{2\pi l \epsilon_0}\int_a^b \frac{1}{r}\,dr$$

$$\Rightarrow V = (Q/2\pi l\epsilon_0)\ln(b/a) \Rightarrow C = Q/V = 2\pi l\epsilon_0/\ln(b/a)$$

Thus:

$$\boxed{C = 2\pi l\epsilon_0/\ln(b/a)}$$

→ Once again we can readily see that capacitance only depends on geometry of the conductor arrangement.

Concentric Spherical Shell Capacitor:

Here the electric field in the region between conductors (by shell theorem):

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}\hat{r}.$$ Thus V:

$$V = \int_a^b \vec{E}\cdot d\vec{r} = \frac{Q}{4\pi\epsilon_0}\int_a^b \frac{1}{r^2}\,dr$$

$$\Rightarrow V = \frac{Q}{4\pi\epsilon_0}\left[\frac{b-a}{ab}\right]$$

$$\Rightarrow C = \frac{Q}{V} = 4\pi\epsilon_0 ab/(b-a),\ \text{Thus}\ \boxed{C = \frac{4\pi\epsilon_0 ab}{b-a}}$$

→ Note that Electrostatic Energy stored between radii a and b is simply $U = Q^2/2C$, for all other regions, energy is zero. Energy stored between certain radii due to a point charge can always be found by modelling the region by concentric spherical shell capacitor which generates the same electric field.

YOUTUBE.COM/@THEPHYSICSSHELTER

Capacitance Of Isolated Spherical Shell:

Recall for an isolated conductor, for purpose for finding capacitance, the other conductor is assumed to be at infinity. Thus in the result for concentric shell capacitor we can make $b \to \infty$ and $a = R$ (∵ shell radius is assumed R). Now $C = \frac{4\pi\epsilon_0 ab}{(b-a)}$ with our substitutions it becomes:

$$\boxed{C = 4\pi\epsilon_0 R}$$

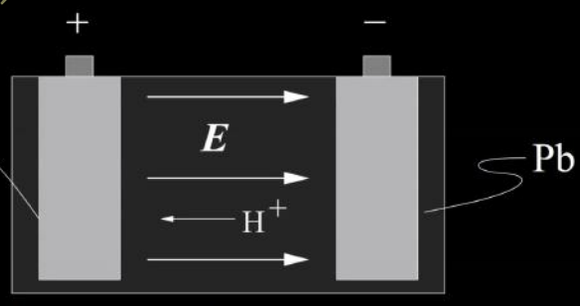
(Same result also holds true for solid conducting sphere)

Finding Self Energy Of Conducting Sphere Using Capacitance:

Recall that $U = Q^2/2C = \frac{Q^2}{2(4\pi\epsilon_0 R)} = \frac{kQ^2}{2R}$, this is the same result that we obtained during our study of electrostatics.

Battery As A Circuit Element:

The purpose of a battery can be likened to that of a pump in water circuits. A battery pumps charges from a low potential energy to a higher potential energy using various mechanisms. (Could be by chemical mechanism, magnetic mechanism or some other mechanism. Inside a battery this increase in potential energy is achieved by making positive charges flow against the electric field or negative charges along the field. This energy comes at expense of other energies in the battery (for instance chemical energy in a lead acid battery).



$$Pb + HSO_4^- \rightarrow PbSO_4 + H^+ + 2e^-$$

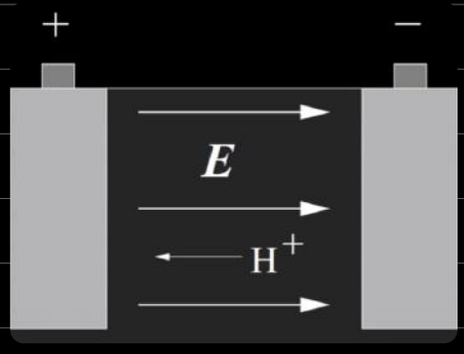
$$PbO_2 + 3H^+ + HSO_4^- + 2e^- \rightarrow PbSO_4 + 2H_2O$$

An ideal battery has zero internal resistance and maintains a constant potential difference. This potential difference can be used to make current flow through external circuit elements (also charging capacitors).

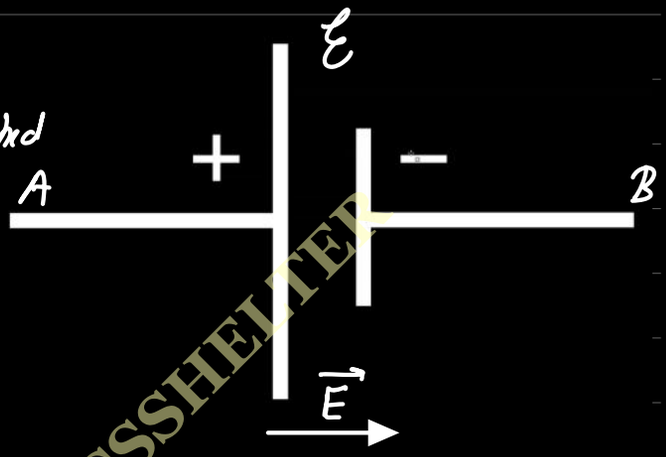
YOUTUBE.COM/@THEPHYSICSSHELTER

EMF OF A BATTERY: Work done by the non conservative forces (say chemical forces etc) in pushing the charges against electric field inside the battery per unit charge is called the EMF of the battery. Mathematically,

$$\mathcal{E} = W_{bat}/q$$



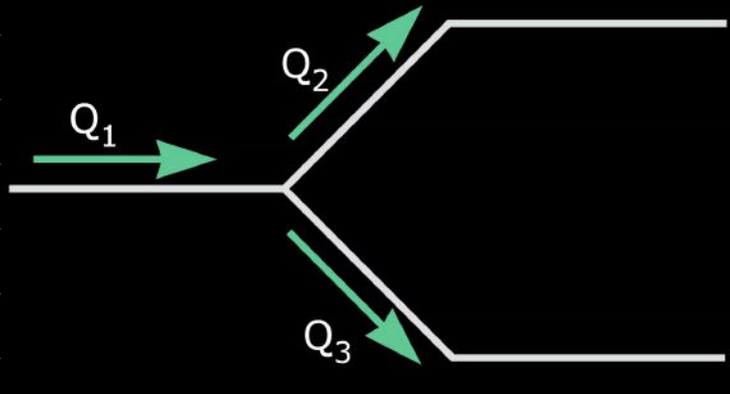
The adjacent picture shows the symbol for an ideal cell. If a charge is pumped from negative terminal to the positive terminal, the battery is said to have done positive work, i.e if a charge $+q$ flows from B to A passing through the battery, work done by battery is $+q\mathcal{E}$, while if a charge $+q$ passes from A to B from within the battery, the work done by the battery is taken as $W = -q\mathcal{E}$. An ideal battery always maintains a constant potential difference across its terminals (equal to its EMF). Whenever a battery is connected in a circuit, the amount of charge given out by one terminal ultimately returns to the other terminal after passing through external circuit.



KIRCHOFF'S LAWS: For solving charge flows in circuits, there are certain logical rules to be followed while considering various junctions and branches in circuits, these rules are called Kirchoff's laws.

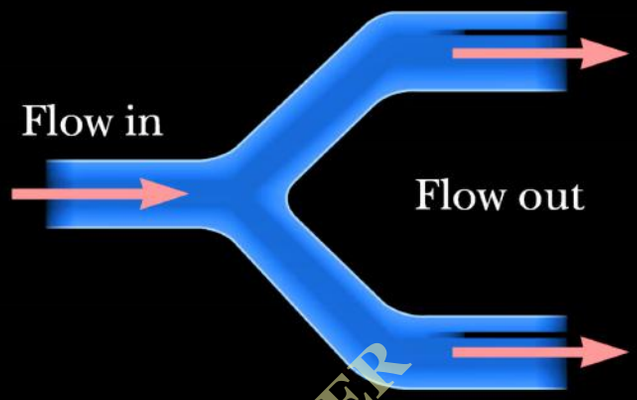
KIRCHOFF'S FIRST RULE (JUNCTION LAW)

In steady state (if there is no accumulation of charge at a junction), the net charge flowing in to a junction is equal to amount of charge flowing out of it. (Or Algebraic sum of all charges flowing in to a junction is zero).



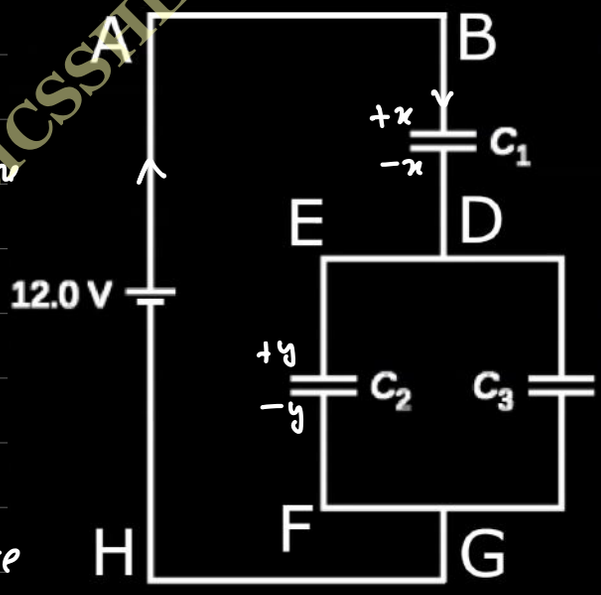
YOUTUBE.COM/@THEPHYSICSSHELTER

Thus $Q_1 = Q_2 + Q_3$ (According to first statement) or
$Q_1 - Q_2 - Q_3 = 0$ (According to second statement).
Here we counted charge flowing in to junction as positive and the charge leaving the junction as negative).
This law is based on conservation of charges. Just as flow entering a joint must be equal to outflow in absence of accumulation, in the same way first rule is a continuity equation for charge flow.



Kirchoff's Loop Rule (Voltage Law):

Summation of potential drops across all elements in a closed loop is zero. This is true because electrostatic field is a conservative field, therefore in any closed loop $\int \vec{E} \cdot d\vec{l} = 0$ and $\int \vec{E} \cdot d\vec{l}$ across any element of the circuit element represents potential drop across it.



→ While traversing along a loop if we hit negative terminal of a battery of EMF $\mathcal{E}$, we should write a $-\mathcal{E}$ for the potential drop while crossing it, similarly if we hit positive terminal, we should write $+\mathcal{E}$ for potential drop.

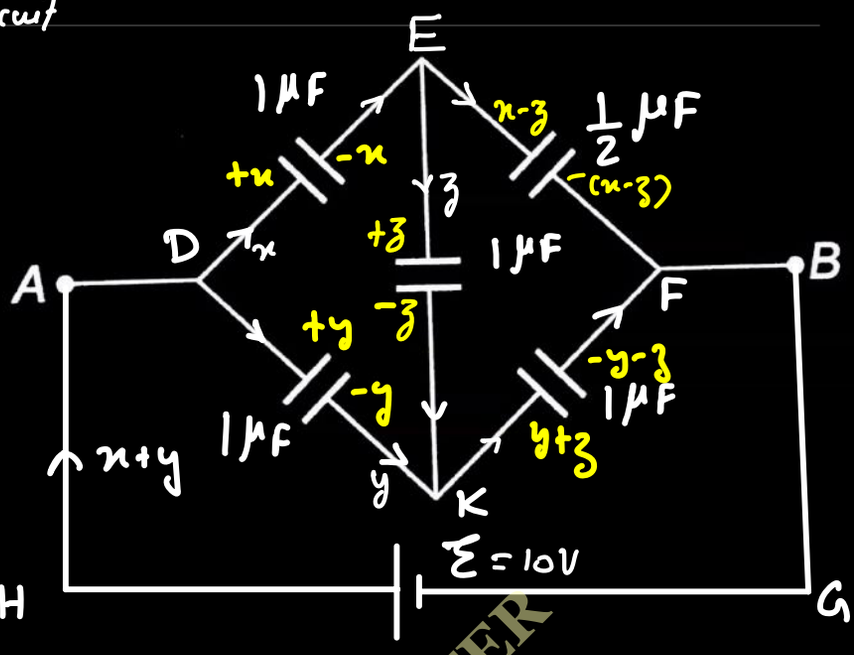
→ While traversing a loop if we hit a capacitor plate having a charge $+q$, we should write $+q/C$ for potential drop across capacitor (where C is capacitance), similarly if we hit negative plate of the capacitor, we must write $-q/C$ for potential drop encountered while crossing the capacitor.

Example: Write loop law equation for the loop ABDEFGHA in the circuit shown above.

Solution: Charge variables are marked on the capacitors, using the understanding from loop law we can write: $\frac{x}{C_1} + \frac{y}{C_2} - 12 = 0$ Ans.

YOUTUBE.COM/@THEPHYSICSSHELTER

EXAMPLE: Consider the capacitor circuit shown in the figure. Each capacitor is 1 μF (microfarad), except one which is 1/2 μF. Find the charge variables x, y and z flowing through various branches. (x, y, z represent charges in microcoulombs).



Solution: For three unknowns we will need three independent equations this can be done by considering three different loops:

Loop DEKD: $\frac{x}{1} + \frac{z}{1} - \frac{y}{1} = 0$ ①

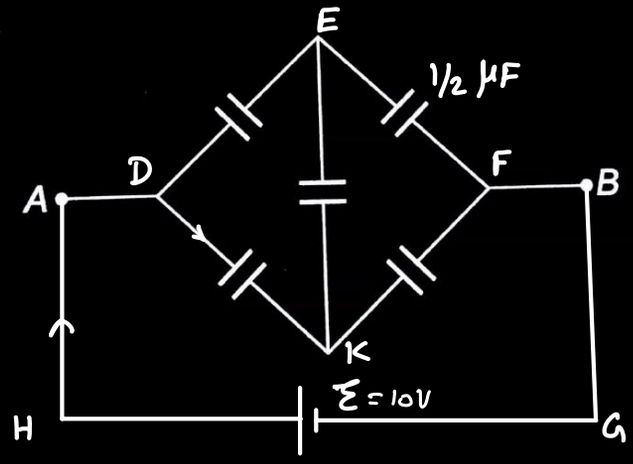
Loop EFKE: $\left(\frac{x-z}{1/2}\right) + \frac{-y-z}{1} \quad \frac{-z}{1} = 0$ ②

Loop ADKFBGHA: $\frac{y}{1} + \frac{y+z}{1} - 10 = 0$ ③

Solving ①, ② and ③: $x = 50/13$, $y = 60/13$, $z = 10/13$. (All charges in microcoulomb's). Now we can find charges on each capacitor individually. For example charge on 1/2 μF capacitor:

$q = x - z = 40/13$ μC.

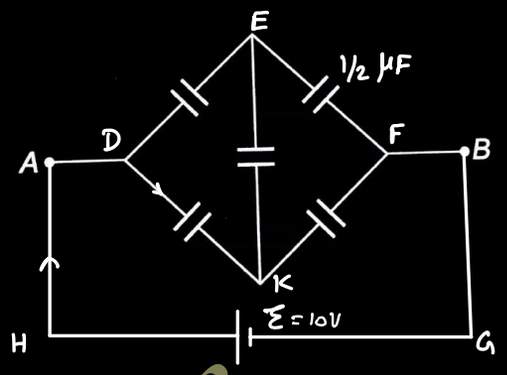
NODE VOLTAGE METHOD: In this method we assign voltage (Potential) variables to various nodes (junctions) in the circuit and solve for the node potentials by applying junction rule to various junctions. Since the charge flows depend on potential differences rather than absolute potentials, thus at one of the nodes we can assign a zero potential (call it reference node) and all other potentials are defined relative to this



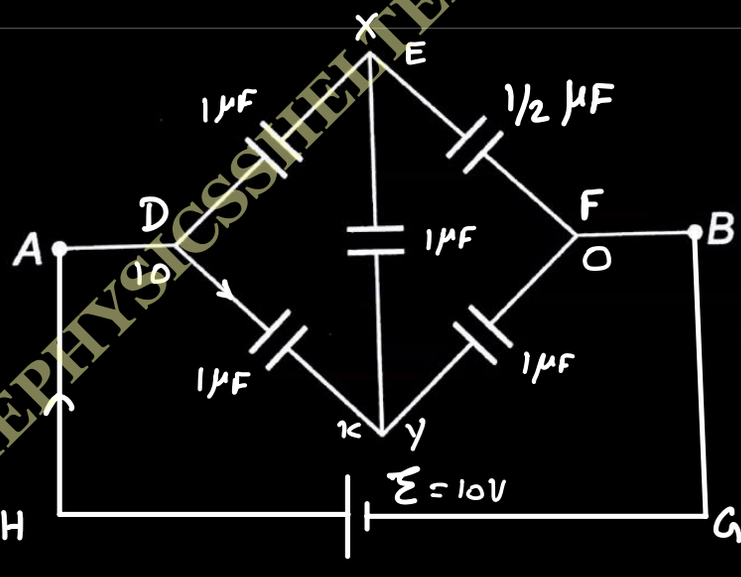
YOUTUBE.COM/@THEPHYSICSSHELTER

node. (Usually it is convenient to consider negative terminal of a battery to be zero potential, though it is not necessary). The method will be clearly understood from following illustration.

EXAMPLE: (considering zero potential at point G in the capacitor circuit, find the potentials of points E and K and hence find the charge on 1/2 μF capacitor (all other capacitors are 1μF each).



Solution: Recall that if V is potential difference across a capacitor, then its charge is CV. Now here we have considered zero potential at point G, Potential of F and B must also be zero (lying on same zero resistance wire). Since battery has EMF of 10 Volt, potential of H, A and D is also 10 Volt each. Now we assign potential variables X and Y to nodes E & K Respectively. Now applying junction law at node E, charge flows along $\vec{ED}$, $\vec{EK}$ and $\vec{EF}$ add to zero thus:



Node E: $(X-10)\times 1 + (X-Y)\times 1 + (X-0)\times 1/2 = 0$ ①

Similarly at node K, net inflow of charge along $\vec{DK}$, $\vec{EK}$ and $\vec{FK}$ must be zero, thus we have the junction rule equation:

$$(10-Y)\times 1 + (X-Y)\times 1 + (0-Y)\times 1 = 0 \quad ②$$

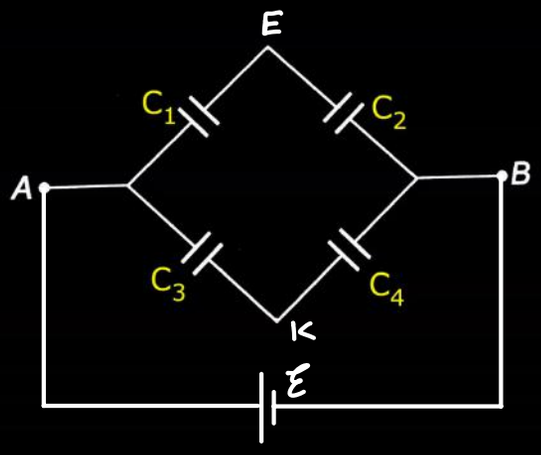
Solving ① and ② for X and Y: $X = 80/13$ Volts and $Y = 70/13$ Volts Ans

Now charge on 1/2 μF capacitor: $q = (X-0)\times\frac{1}{2} = 40/13\ \mu C$ Ans.

(This is the same result that we got earlier using charge variables and applying loop law to various loops)

YOUTUBE.COM/@THEPHYSICSSHELTER

EXAMPLE: Assuming zero potential at node K, find the potential of node E. For what relationship between C_1, C_2, C_3 and C_4 are nodes E and K equipotential.

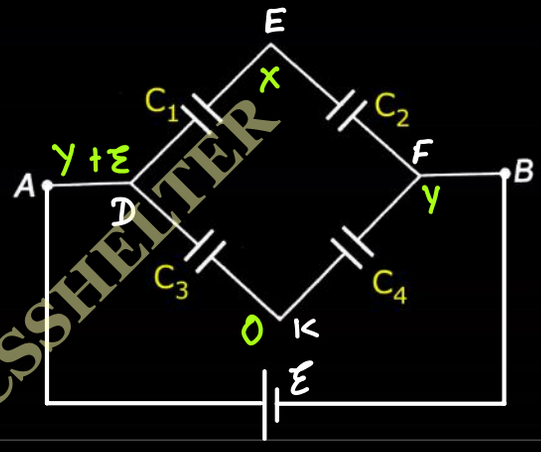


Solution: We assume zero potential at K and negative terminal of battery as Y thus potentials at F and D become Y and $Y+\mathcal{E}$ respectively. Now we assume potential X at node E. Now Junction Rule at E:

$$[X-(Y+\mathcal{E})]C_1 + (X-Y)\times C_2 = 0 \quad (1)$$

Similarly at node K:

$$(Y+\mathcal{E}-0)\times C_3 + (Y-0)\times C_4 = 0 \quad (2)$$



Solving (1) & (2):

$$X = \frac{C_1C_4 - C_2C_3}{(C_1+C_2)(C_3+C_4)} \quad \text{and} \quad Y = \frac{-C_3\mathcal{E}}{C_3+C_4}$$

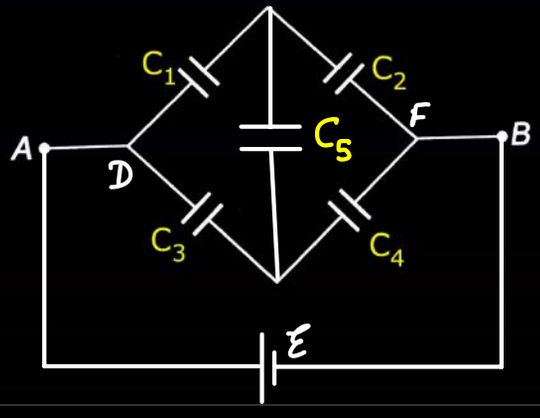
Now for E & K to be equipotential, $X=0$

$$\Rightarrow \quad C_1C_4 - C_2C_3 = 0$$

$$\Rightarrow \quad \frac{C_1}{C_3} = \frac{C_2}{C_4} \quad \text{or} \quad \frac{C_1}{C_2} = \frac{C_3}{C_4},$$

BALANCED WHEATSTONE BRIDGE:

In the circuit shown in the adjacent figure, if $\frac{C_1}{C_2} = \frac{C_3}{C_4}$ then no charge flows in to C_5 irrespective of its value, this situation is known as balanced wheatstone bridge (originally the concept was used in resistor circuits). This has a simple justification,



we saw in our analysis that if $C_1/C_2 = C_3/C_4$ then top node and bottom node become equipotential and hence even if a capacitor is connected across them, no charge flows through it.

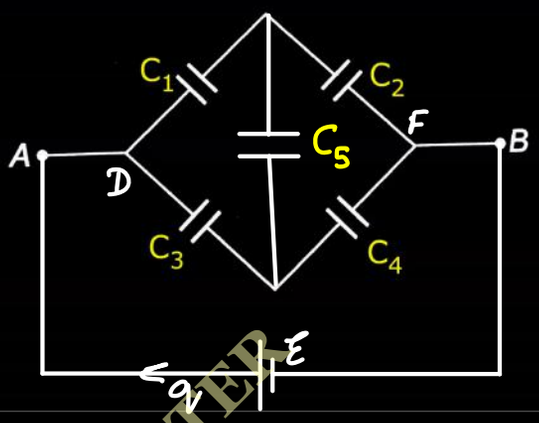
YOUTUBE.COM/@THEPHYSICSSHELTER

(Junction laws and loop laws are already satisfied without any need for charge flow through C_5).

Effective Capacitance Of A Network:

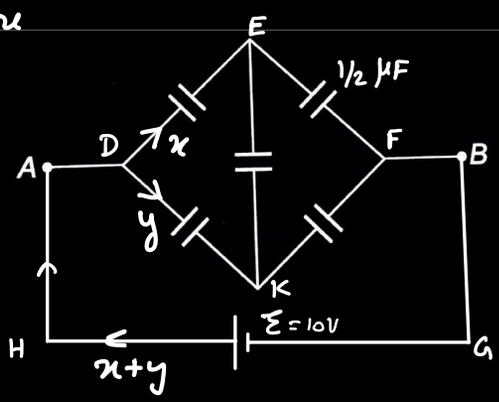
If q is the charge drawn by a network of capacitors from a battery of EMF $\mathcal{E}$, the effective capacitance of the network is said to be simply:

$$C_{eff} = q/\mathcal{E}$$



→ Thus for finding effective capacitance of any circuit, we can connect the network to 1V Battery and the find the charge flown through it.

Example: In the shown network all capacitors are 1 μF each except the one marked 1/2 μF. Find the effective capacitance between points A and B.

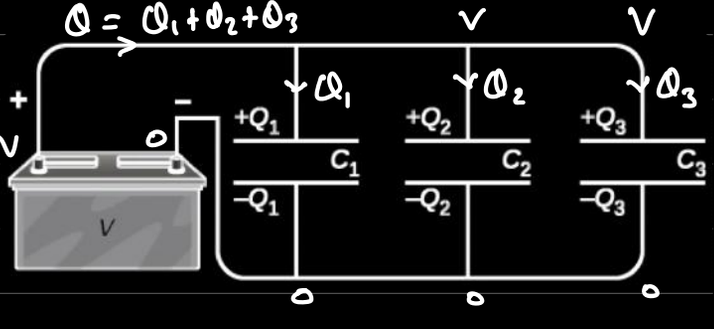


Solution: This is the same network that we had earlier solved using loop law and node voltage. From our earlier analysis, we got $x = 50/13\ \mu C$ and $y = 60/13\ \mu C$. Thus net charge flown from the battery $q = x+y = 110/13\ \mu C$. Now since $\mathcal{E} = 10\ V$, $C_{eff} = q/\mathcal{E} = 11/13\ \mu F$ Ans.

Parallel Combination Of Capacitors: Two or more capacitors are said to be connected in parallel if they have same potential difference by virtue of configuration (or connections).

For instance in adjacent figure, all capacitors have same potential difference V. Here since potential drop across each capacitor is V, the individual charges are given as:

$$Q_1 = C_1 V,\quad Q_2 = C_2 V,\quad Q_3 = C_3 V$$



YOUTUBE.COM/@THEPHYSICSSHELTER

Now total charge flown from battery $Q = Q_1 + Q_2 + Q_3 = V(C_1 + C_2 + C_3)$

Thus $C_{eff} = Q/V$ or:

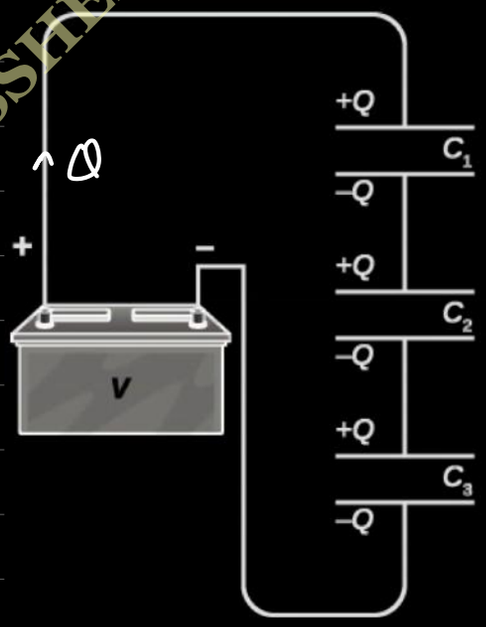
$$\boxed{C_{eff} = C_1 + C_2 + C_3}$$

Here it can be readily seen that charge on any capacitor is directly proportional to its capacitance, thus if Q is net charge flown in to a parallel combination, individual charges can be obtained by dividing the total charge in ratio of capacitances thus:

$$\boxed{Q_1 = \frac{C_1}{C_1 + C_2 + C_3} Q}$$

Similarly $Q_2 = Q(C_2/C_1 + C_2 + C_3)$ etc.

CAPACITORS IN SERIES: Two or more capacitors are said to be connected in series if they carry same charge by virtue of configuration (they are on same branch), such as the configuration shown in the figure. Now applying loop law to the loop,



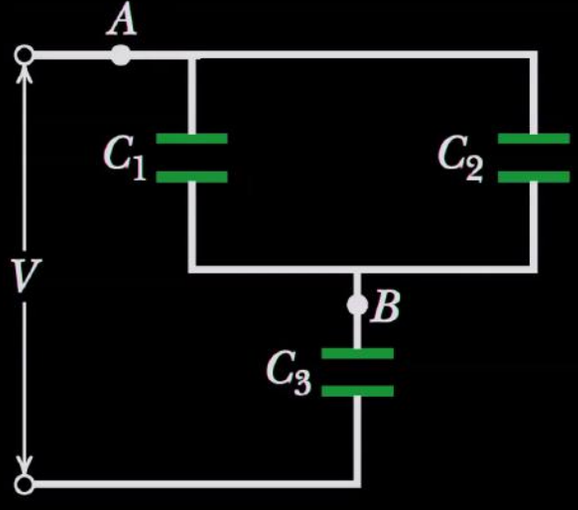
$$\frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3} - V = 0$$

$$\Rightarrow \frac{V}{Q} = \frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$, thus for parallel combinations: $\boxed{\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}}$

EXAMPLE: In the adjacent figure, a potential difference $V = 120$ Volts is applied across the capacitor network. Given, $C_1 = 4\ \mu F$, $C_2 = 2\ \mu F$ and $C_3 = 3\ \mu F$. Find:

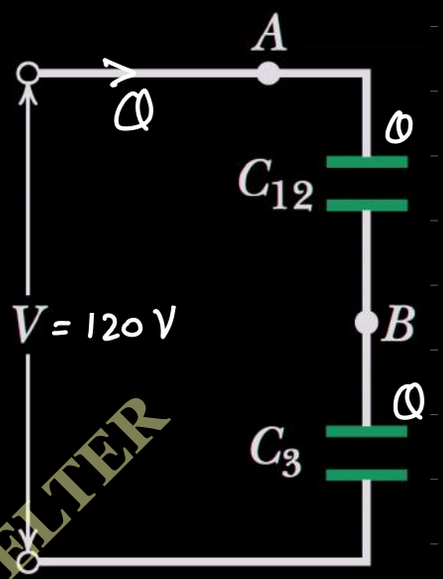
(a) Effective Capacitance of combination.

(b) Individual charges on C_1, C_2 and C_3.



YOUTUBE.COM/@THEPHYSICSSHELTER

Solution: The effective capacitance can be found by seeing that C_1 is ∥ to C_2 and the combination of C_1 and C_2 is in series with C_3.
Now effective capacitance of C_1 & C_2, $C_{12} = C_1 + C_2 = (2+4)\mu F = 6\mu F$



Now C_{12} is in series with C_3 thus,

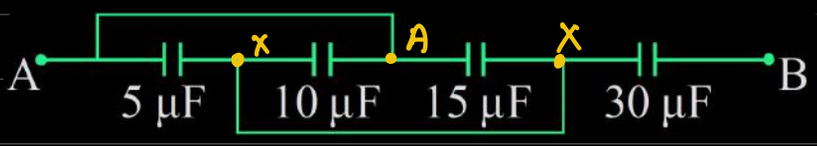
$$\frac{1}{C_{eff}} = \frac{1}{C_3} + \frac{1}{C_{12}} \Rightarrow C_{eff} = \frac{C_3 \times C_{12}}{C_3 + C_{12}} = \frac{6\times 3}{6+3}\mu F$$

$\Rightarrow C_{eff} = 2\mu F$ Ans.

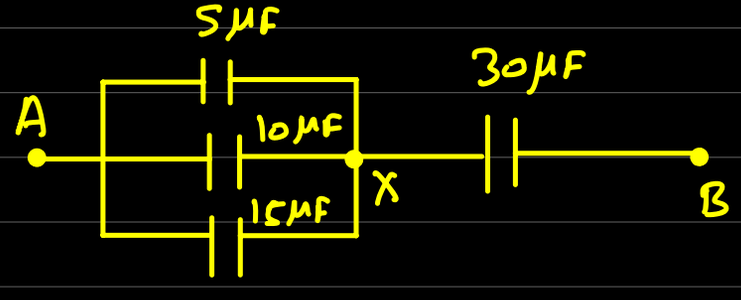
Now total charge drawn from battery, $Q = C_{eff} V$
$\Rightarrow Q = 120 \times 2 = 240\mu C$. Thus net charge on $C_1 \| C_2 = 240\mu C$, also Q on $C_3 = 240\mu C$ Ans.
Now in parallel, the charge is divided in ratio of capacitances thus: $Q_1 = \left(\frac{C_1}{C_1+C_2}\right)Q = \frac{4}{6}\times 240 = 160\mu C$ Ans.

similarly, $Q_2 = \left(\frac{C_2}{C_1+C_2}\right)Q = \frac{2}{6}\times 240 = 80\mu C$ Ans

Simplifying Capacitor Circuits Using Node Potentials:



Sometimes a complicated looking circuit might be greatly simplified by redrawing the circuit in such a way that potential difference across each element remains same as before, however a unique potential is indicated by a single point. For instance in the above circuit, the point marked A in orange has same potential as left most point, similarly points marked X have same potential.
Thus we observe:
5μF is connected between A and X
10μF is also connected between A and X
15μF is also connected between A and X
30μF is connected between B and X



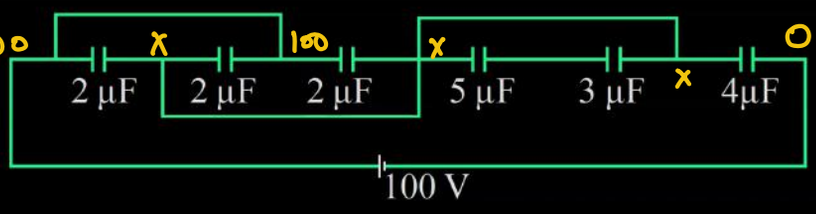
Using this information, simplified diagram is shown above, and C_{eff}:

$$(5+10+15)\text{Series}(30\,\mu F) = \frac{30\times 30}{30+30} = 15\,\mu F \text{ Ans.}$$

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

EXAMPLE:

Find the charge on 5μF capacitor in the shown circuit.



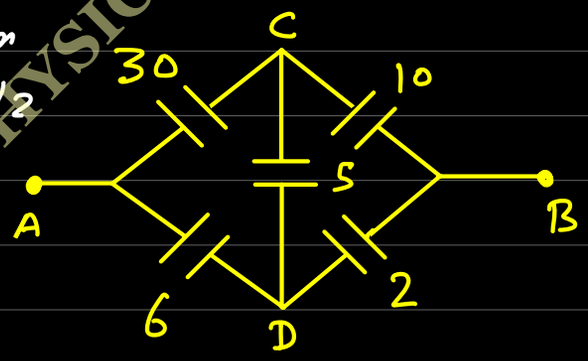
Solution: Here it is immediately recognised that potential difference across the series combination of 5μF and 3μF is zero, thus no charge flows in this branch thus our required answer: 0μC Ans.

EXAMPLE: In the shown adjacent circuit, all capacitances are marked is μF. Find effective capacitance between A & B.

Hint: Spot the balanced Wheatstone Bridge.

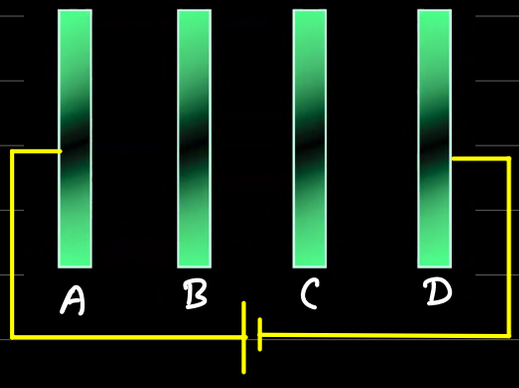
Solution: Simplified circuit is drawn in adjacent figure. Recognise that 30/6 = 10/2 Thus it is a balanced wheatstone bridge thus making 5μF capacitor redundant. Now capacitance is readily calculated using series / parallel simplification.

$$C_{eff} = \left(\frac{30\times10}{30+10}\right) + \left(\frac{6\times2}{6+2}\right) = 9\,\mu F \text{ Ans.}$$



FINDING CAPACITANCE OF PARALLEL PLATE SYSTEMS USING TRIFURCATION TECHNIQUE

Sometimes we are given a system of neutral parallel plates being connected across a battery. In such a system since net charge of the system is zero, extreme faces get zero charges (Recall the parallel plate theory from electrostatics), also facing faces get equal and opposite charges, thus each air gap between plates serves as a capacitor and metal matrix between two sides of any plate serves as conducting wire. For clarity consider the following diagram. Here left and right surfaces of plate B are marked B^- and B^+, and the region between dotted lines can be



YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

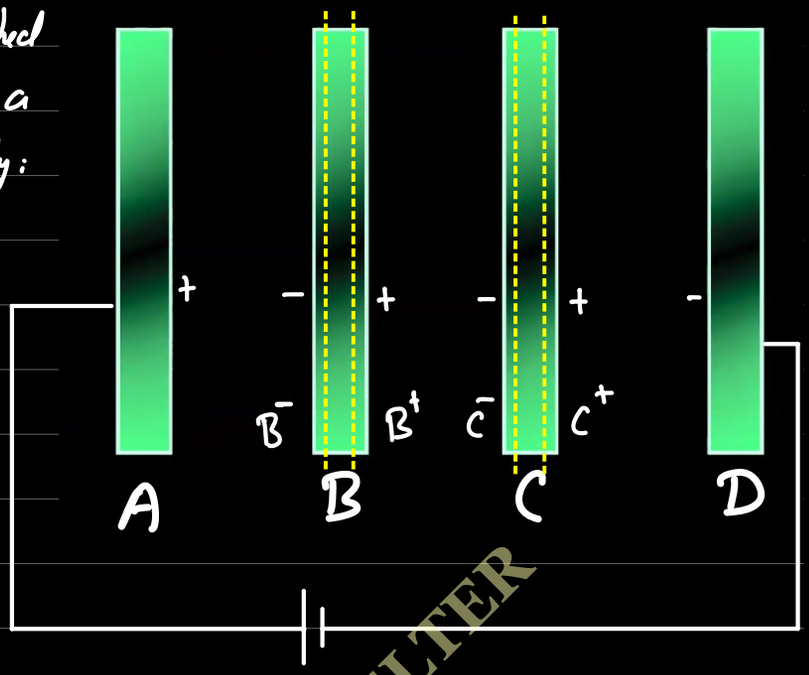
considered as a connecting wire between the conducting surfaces marked B_- and B_+. Thus now we have a system of three capacitors namely:

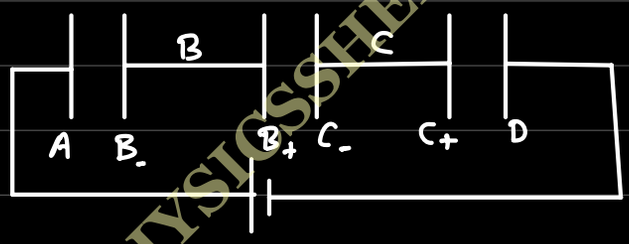
① Between A and B_-

② Between B_+ and C_-

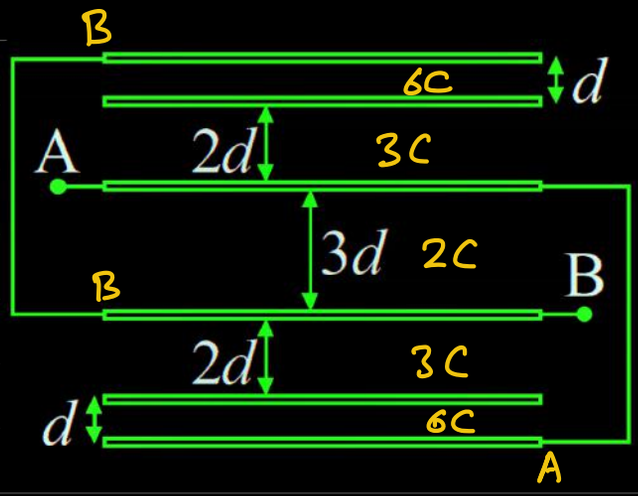
③ Between C_+ and D.

Now we can redraw the simplified equivalent circuit to find the net effective capacitance. Now if A is each plate area and d is the air gap, then each capacitor's capacitance is $C = \epsilon_0 A/d$ and effective capacitance due to series is $C_{eff} = C/3 = \epsilon_0 A/3d$ Ans.

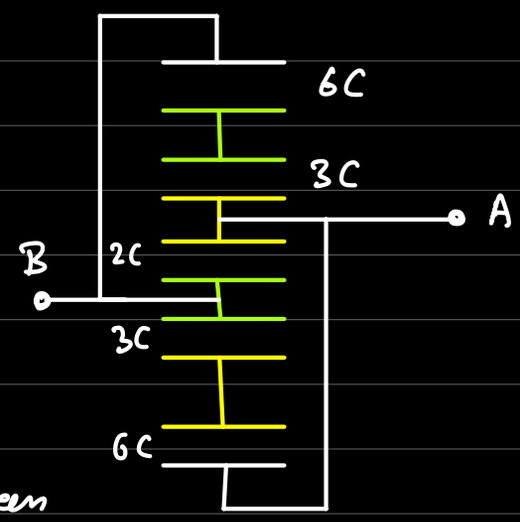




EXAMPLE: Find effective capacitance of shown system between points A and B.



Solution: Let $C = \epsilon_0 A/6d$. Now respective capacitances across every air gap is shown in orange colour. Now we can draw the capacitor circuit as shown in the next figure. (Here the second, third, fourth and fifth plate from the top are trifurcated). Now we can readily see that the circuit is reduced to simple series parallel combination:



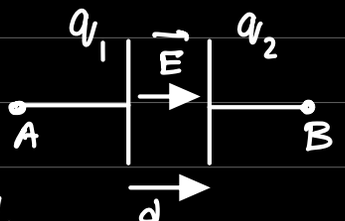
$(6C \text{ Ser } 3C) \,||\, 2C \,||\, (6C \text{ Ser } 3C)$

Since all three branches are connected between the potential difference across A & B. Thus net capacitance of the circuit,

$$C_{eff} = 2C + 2C + 2C = 6C = 6 \times \frac{\epsilon_0 A}{6d} = \epsilon_0 A/d \text{ Ans.}$$

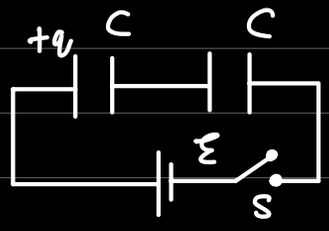
YOUTUBE.COM/@THEPHYSICSSHELTER

Capacitor With Unequal Magnitude Charges As A Circuit Element :

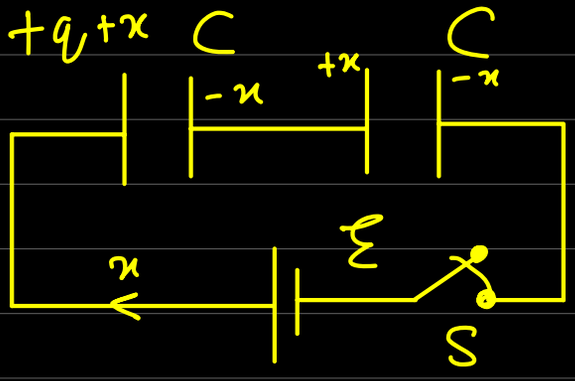


Consider a capacitor having charges q_1 and q_2 on its plates. The electric field in the region between plates is given as $E = \frac{q_1}{2A\epsilon_0} - \frac{q_2}{2A\epsilon_0}$ accordingly as one moves from left plate to right plate, the potential drop encountered $V = Ed = \left(\frac{q_1 - q_2}{2}\right)\frac{d}{\epsilon_0 A} = \frac{q_1 - q_2}{2C}$, thus for such a capacitor, $\boxed{V = \frac{q_1 - q_2}{2C}}$

Example: Consider the circuit shown in the adjacent figure. The left plate of one of capacitors is already given a charge q. Find the charge supplied by the battery when the switch is closed.



Solution: Assuming that battery supplies a charge x, the charges on various capacitor plates are shown in adjacent figure. Now applying loop law:



$$\frac{q + 2x}{2C} + \frac{x}{C} - \mathcal{E} = 0$$

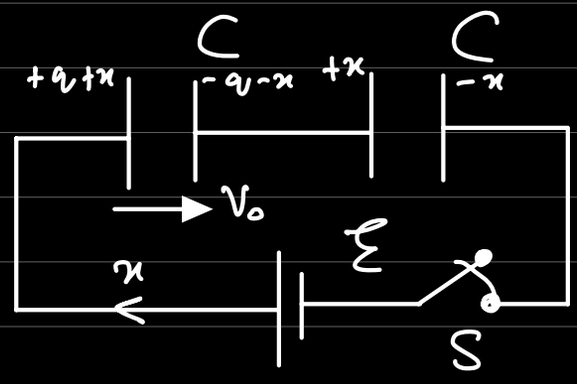
$$\Rightarrow \quad x = \frac{C\mathcal{E}}{2} - q/4 \quad \text{Ans.}$$

→ Once we know the value of x, we can find the individual charges on plates also.

YOUTUBE.COM/@THEPHYSICSSHELTER

Precharged Capacitor As Superposition Of Battery And Uncharged Capacitor

Consider the circuit shown in the adjacent figure in which the left capacitor is already charged to a potential V_0 (such that $q/C = V_0$) and now we close the switch and we are interested in finding additional charge flown from the battery in the process.

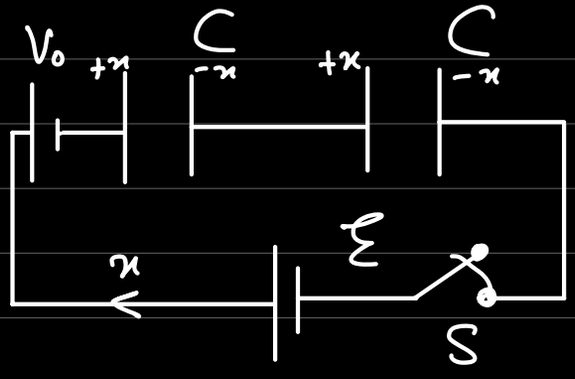


Let the charge flown be x, then applying loop law we get:

$$\frac{q+x}{C} + \frac{x}{C} - \mathcal{E} = 0 \Rightarrow \frac{q}{C} + \frac{2x}{C} - \mathcal{E} = 0$$

$$\Rightarrow V_0 + 2x/C - \mathcal{E} = 0 \quad (1)$$

Now suppose we have an alternate circuit: Here instead of a capacitor precharged to V_0, we have an uncharged capacitor and a battery of EMF V_0. Now let us apply loop law to this circuit:



$$V_0 + \frac{x}{C} + \frac{x}{C} - \mathcal{E} = 0$$

$$\Rightarrow V_0 + 2x/C - \mathcal{E} = 0 \quad (2)$$

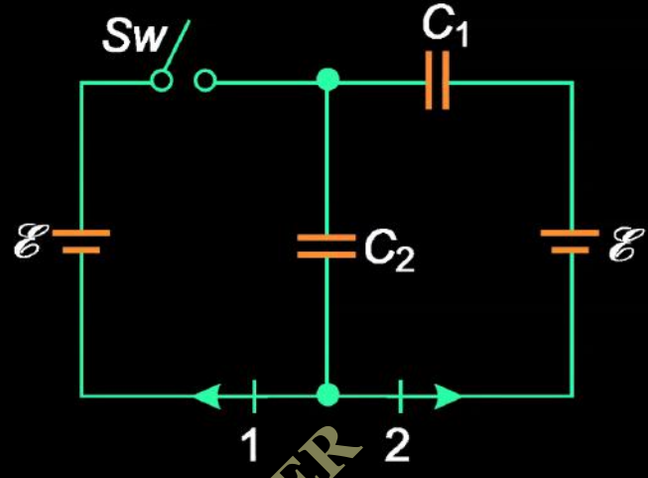
From (1) and (2) we see that we get identical loop equations for both circuits, thus for the purpose of calculating additional charge flows in any branch, we can break the charged capacitor in to a battery and an uncharged capacitor.

→ For finding net final charge on precharged capacitor remember to add the original charge to the additional charge flow after switch closing.

YOUTUBE.COM/@THEPHYSICSSHELTER

Heat Generation In Capacitor Circuits:

Whenever a switch is shifted in a batteries and capacitors network, sudden flow of charge happens and the charges on capacitors change almost instantaneously. Whenever charges accelerate, they emit electromagnetic radiation, thus during switching process, some heat is lost to the atmosphere. This loss can be estimated by applying First Law of Thermodynamics to capacitor circuits i.e:



$$\boxed{\Delta W_{bat} = \Delta U + \Delta H}$$

Here ΔW_{bat} = Work done by batteries

ΔU = Change in stored energy in Capacitors

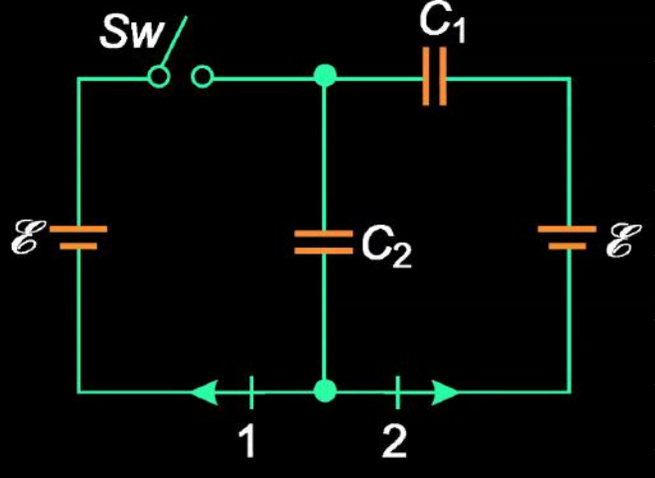
ΔH = Heat lost to atmosphere.

$$\Rightarrow \boxed{\Delta H = \Delta W_{bat} - \Delta U}$$

→ For estimation of ΔW_{bat}, we will need to find the charges flown through battery which in turn can be found by tracing the charge through all the distributaries of the battery branch until they finally end at some capacitor plate. Then total charge flown can be found by adding up the change in charges at terminal plates of all distributaries. This will be clear from a following example.

→ Alternatively the charge flown through battery can also be found by tracing the tributaries until we reach the source plate, now the charge flow through battery must be equal to loss of charge on all the source plates.

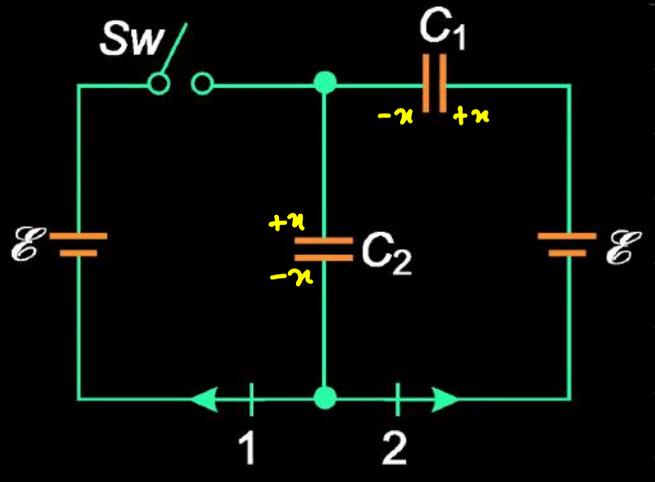
Example: Irodov 3.122: What charges will flow through sections ① and ② in the shown circuit after switch is closed. Also calculate the work done by batteries and heat generated after the switching process.



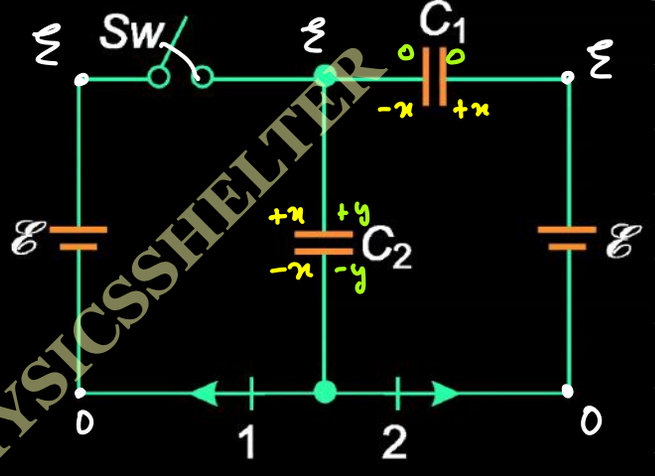
YOUTUBE.COM/@THEPHYSICSSHELTER

<u>Solution:</u> For finding charge flows we will need initial and final charges on capacitor plates. Initially C_1 & C_2 are in series thus their charge is given as:

$$x = \frac{C_1 C_2}{C_1 + C_2}\,\mathcal{E}$$



Now after switch is closed, using node potentials it can be readily seen that potential difference across C_1 becomes zero, thus final plate charge of C_1 is zero. Now using node potentials, final charge on $C_2 = y = \mathcal{E}C_2$.



Now tracing distributaries of section ①, whatever flows along section ① must either accumulate in either left plate of C_1 or top plate of C_2. Thus, charge flown through section ①:

$$q_1 = y - x + 0 - (-x) = y = \mathcal{E}C_2 \quad \text{Ans.}$$

Similarly charge flown from section ② accumulates on right plate of C_1
Thus charge flown from ② $q_2 = 0 - x = -x = -[C_1C_2/(C_1+C_2)]\,\mathcal{E}$ Ans

Now Total work done by Batteries, $\Delta W_{bat} = \mathcal{E}q_1 + \mathcal{E}q_2$

$$\Rightarrow \Delta W_{bat} = \mathcal{E}^2\left[C_2 - \frac{C_1C_2}{C_1+C_2}\right] = \frac{\mathcal{E}^2 C_2^2}{C_1+C_2} \quad \text{Ans.}$$

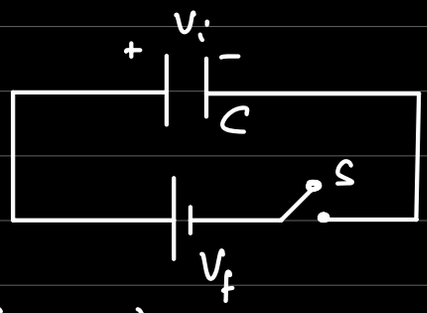
Now $U_i = \frac{1}{2}\left(\frac{C_1C_2}{C_1+C_2}\right)\mathcal{E}^2$, $U_f = \frac{1}{2}C_2\mathcal{E}^2 \Rightarrow \Delta U = \frac{1}{2}\mathcal{E}^2\left[C_2 - \frac{C_1C_2}{C_1+C_2}\right]$

$$\Rightarrow \Delta U = \frac{1}{2}\frac{C_2\mathcal{E}^2}{C_1+C_2}, \quad \text{Now } \Delta H = \Delta W_{bat} - \Delta U = \frac{1}{2}\frac{C_2\mathcal{E}^2}{C_1+C_2} \quad \text{Ans.}$$

YOUTUBE.COM/@THEPHYSICSSHELTER

EXAMPLE: Show that if a capacitor is precharged to a potential V_i when further charged to a potential V_f by connecting to a battery of voltage V_f, then the heat generated is given as $\Delta H = \frac{1}{2}C(V_f - V_i)^2$

Solution: $U_i = \frac{1}{2}CV_i^2 \quad U_f = \frac{1}{2}CV_f^2$



$$\Rightarrow \quad \Delta U = \frac{1}{2}C(V_f^2 - V_i^2)$$

Now charge flown after closing switch, $q = C(V_f - V_i)$

$\Rightarrow \Delta W_{bat} = C(V_f - V_i) \times V_f$. Now $\Delta H = \Delta W_{bat} - \Delta U$

$$\Rightarrow \quad \Delta H = CV_f(V_f - V_i) - \frac{1}{2}C(V_f^2 - V_i^2) = C(V_f - V_i)\left[V_f - \frac{(V_f + V_i)}{2}\right]$$

$$\Rightarrow \quad \boxed{\Delta H = \frac{1}{2}C(V_f - V_i)^2} \text{ Ans.}$$

→ Note that here V_i must be considered as a signed value if V_i has same polarity as V_f across the capacitor then it is considered as positive. If it has opposite polarity, it is to be considered negative.

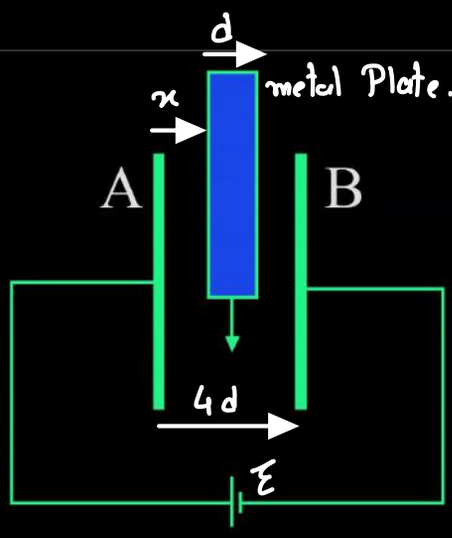
→ If a capacitor is initially uncharged, exactly 50% of the work done is stored as electrostatic energy and rest 50% is lost as heat. (This can be readily seen by putting $V_i = 0$ & $V_f = V$, then $\Delta H = \frac{1}{2}CV^2 = \Delta U = \frac{1}{2}\Delta W_{bat}$).

→ Heat is generated in capacitor circuits only when charges flow suddenly (this causes acceleration of charges), however if charges are made in the circuits very slowly (such as slowly increasing/decreasing separation of plates or slowly inserting or removing a conductor or dielectric plate), no heat is generated. However for applying the first law of thermodynamics we will need to consider work done by external agent bringing about the slow changes in the system. Thus for such cases:

$$\Delta W_{bat} + \Delta W_{ext} = \Delta U \quad (\Delta W_{ext} = \text{Work done by external agent}).$$

YOUTUBE.COM/@THEPHYSICSSHELTER

Example: A capacitor of capacitance C was connected to a battery of EMF $\mathcal{E}$, now a metal plate of thickness d is slowly brought between the capacitor plates using an external agent until plate completely occupies height of the capacitor plates. In this process, find, (a) Work done by battery (b) Work done by external agent.



Solution: Initial capacitance, $C = \epsilon_0 A/4d$, $U_i = \frac{1}{2} C\mathcal{E}^2$

Finally, $\frac{1}{C_{eff}} = \frac{x}{\epsilon_0 A} + \frac{3d-x}{\epsilon_0 A} = \frac{3d}{\epsilon_0 A} = \frac{3}{4C} \Rightarrow C_{eff} = \frac{4C}{3}$

$\Rightarrow U_f = \frac{1}{2} \cdot \frac{4C}{3} \mathcal{E}^2 = \frac{2}{3} C\mathcal{E}^2$

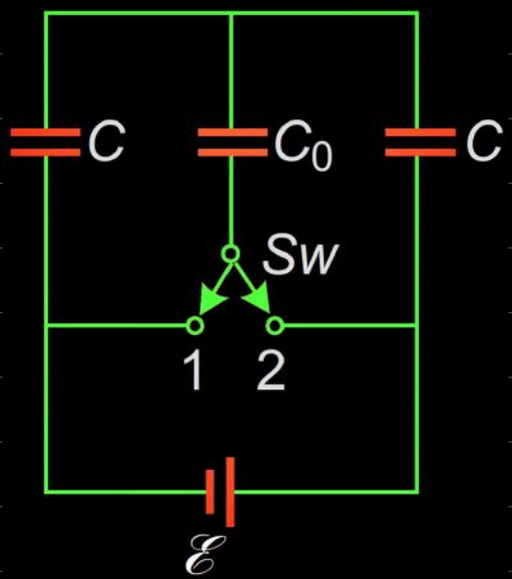
Now $W_{bat} + W_{ext} = \Delta U = \frac{1}{2}\mathcal{E}^2\left[\frac{4C}{3} - C\right] = \frac{1}{6} C\mathcal{E}^2$

Now Work done by battery $\Delta W_{bat} = \left(\frac{4}{3} C\mathcal{E} - C\mathcal{E}\right)\mathcal{E} = \frac{1}{3} C\mathcal{E}^2$ Ans.

$\Rightarrow \Delta W_{ext} = \Delta U - \Delta W_{bat} = \frac{1}{6} C\mathcal{E}^2 - \frac{1}{3} C\mathcal{E}^2 = -\frac{1}{6} C\mathcal{E}^2$ Ans.

(Actually slab will be spontaneously attracted inside capacitor and external agent applies outward force to prevent acceleration of slab).

Example: Irodov 3.131: What amount of heat will be generated in the circuit shown in the figure after the switch Sw is shifted from position 1 to position 2?

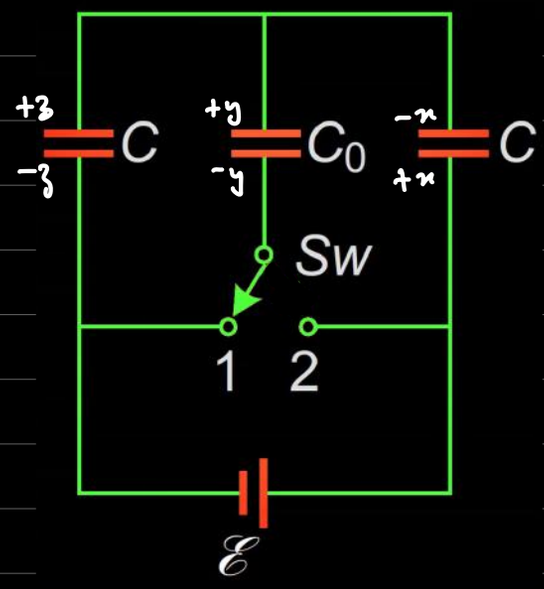


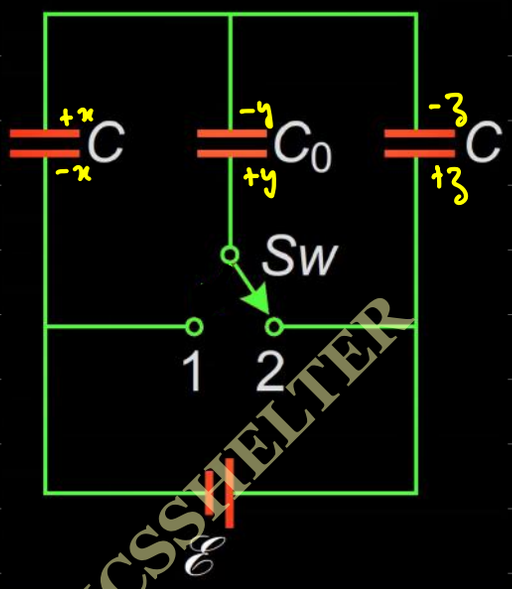
Solution: When switch is in position 1, effective capacitance is seen as the series parallel combination $C_{eff} = (C \| C_0)$ series C and energy stored is $\frac{1}{2} C_{eff} \mathcal{E}^2$.

Now when switch is shifted to position 2, the effective

AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter
YOUTUBE.COM/@THEPHYSICSSHELTER

capacitance is once again the series parallel combination (C||C₀) series C. thus effective capacitance and hence stored energy must still be the same as before. Thus $\Delta U = 0$, which means heat liberated must be equal to the work done by battery.





Initial Condition:

$$x = \frac{(C+C_0)C}{2C+C_0} \times \mathcal{E}$$

$$y = \frac{C_0}{C_0+C} \times x = \frac{C_0 C}{2C+C_0}\mathcal{E}$$

$$z = \frac{C}{C_0+C} \cdot x = \frac{C^2}{2C+C_0}\mathcal{E}$$

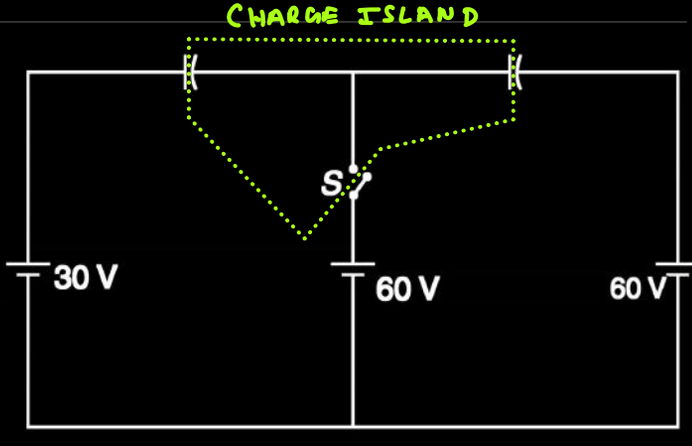
Final Condition

Now charge flown from battery accumulates on lower plates of middle and right capacitor. Thus ΔW_{bat}:

$$\Delta W_{bat} = \mathcal{E}[(z-x) + y - (-y)] = \Delta H$$

$$\Rightarrow \Delta H = \mathcal{E}(z - x + 2y) = \frac{\mathcal{E}^2 C C_0}{2C+C_0} \quad \text{Ans.}$$

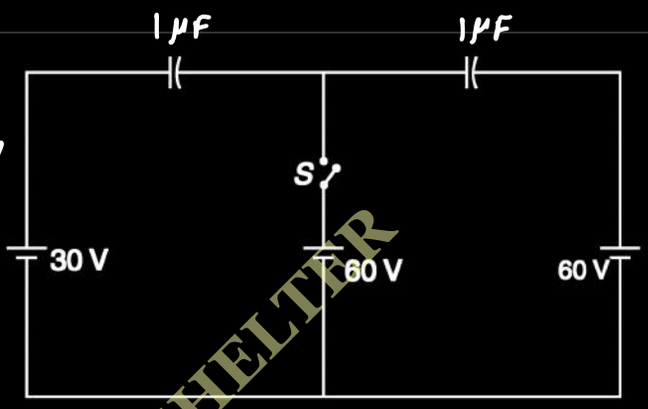
Charge Island: A portion of a circuit not in contact with rest of the circuit through direct conductor contact can be called a charge island if the net charge on this portion is non-zero. For instance if in the adjacent circuit if initially the switch were closed, and then afterwards it is opened, the region inside the dotted boundary is not in direct conducting contact with rest of the circuit and thus its charge will



YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

remain conserved after opening the switch. Whenever such charge islands come in to existence after shifting of switches one must be very careful in charge flow calculations. (The charge conservation constraint must be kept in mind and one can't start the circuit from zero charge for purpose of calculating charge flows).

EXAMPLE: Consider the circuit shown in the figure. Initially the switch was open, and capacitors arranged as shown now the switch is closed



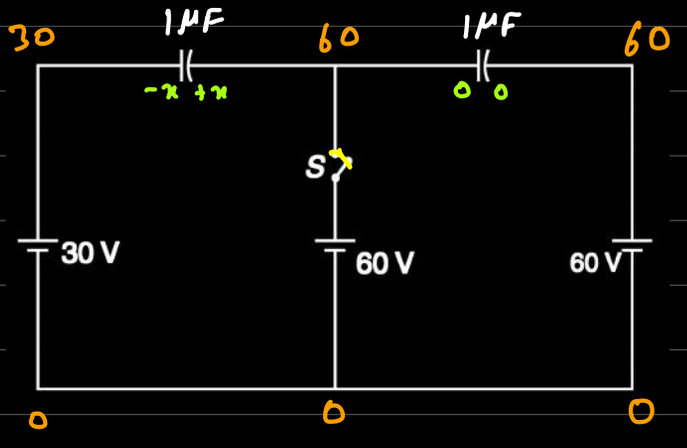
(a) How much charge flowed through the switch.

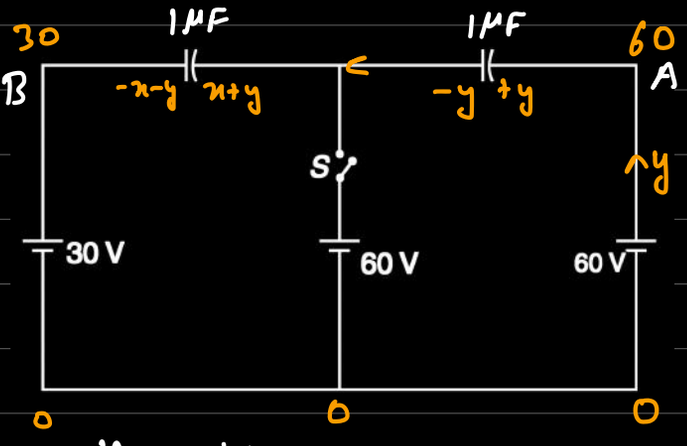
(b) Now if the switch is opened how much charge will flow through 60 V Battery on the right.

Solution: Initially there are no charge islands, the charges on all plates after closing the switch are shown in adjacent figure. (Node Voltage method).

$$x = (60-30) \times 1 = 30\,\mu C.$$

Now when switch S is opened, the charge x becomes trapped in an island. Now let an additional charge y flow through the 60 V battery, Now adding potential drops from A to B:





$$\frac{y}{1} + \frac{x+y}{1} = 60-30 = 30$$

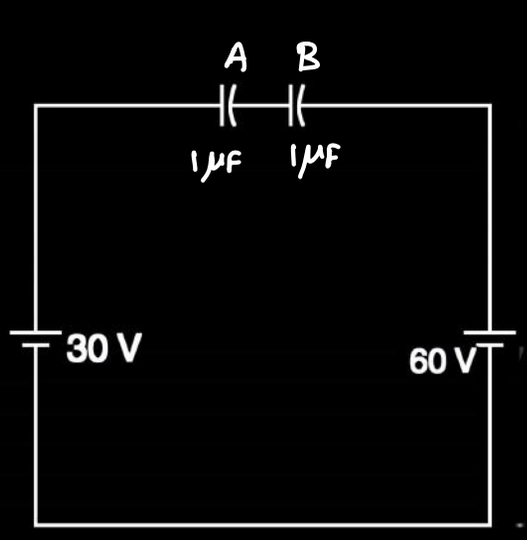
$$\Rightarrow y = \frac{30-x}{2} = \frac{30-30}{2} = 0$$

Thus no additional charge flows upon opening the switch.

→ We could have concluded no charge flow directly be seeing that condition of potential drops is already satisfied without any additional charge flows.

YOUTUBE.COM/@THEPHYSICSSHELTER

Superposition Theorem:



In a system containing only ideal batteries and capacitors, the charge on each capacitor is the superposition of charge due to each battery considered one at a time. (While considering charge due to any battery all other batteries are considered as short circuited.)

For instance in the adjacent circuit, due to 60 V alone, charges on each capacitor is $60 \times \frac{1}{2} = 30\mu C$ with right plates having positive charges.

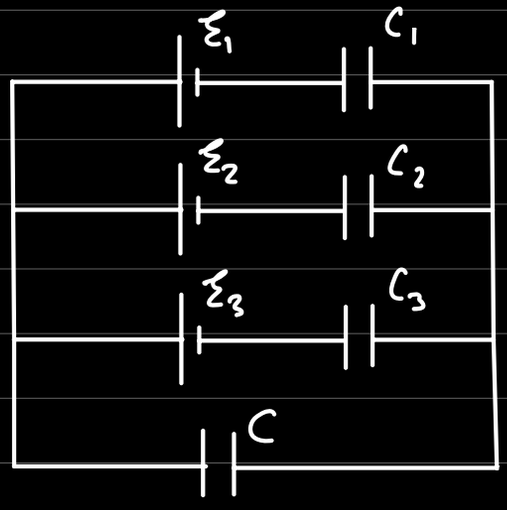
Similarly due to 30 V battery alone, charge on each capacitor will be $30 \times \frac{1}{2} = 15\mu C$ with left plates having positive charges thus by superposition, charge on each capacitor $= 30\mu C - 15\mu C = 15\mu C$ (with right plates positively charged).

→ Superposition theorem holds due to linear nature of systems of equations for loops, thus charge on any capacitor takes the form:

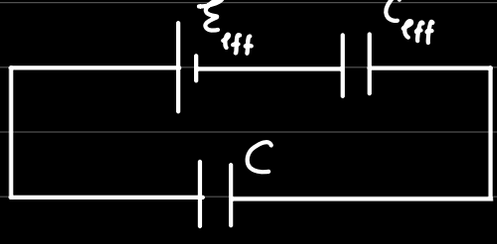
$$q = f(C_i)\,\mathcal{E}_1 + g(C_i)\,\mathcal{E}_2 + \dots \text{ etc.}$$

(There are no terms containing squares or products of EMFs).

Millman's Theorem:



If we have several parallel branches containing batteries and capacitors such as the one shown in adjacent figure, then for purpose of calculating charge flow across the load capacitor C, the parallel branches can be substituted by a single battery and single capacitor, thus circuit reduces to the circuit in the next figure with:



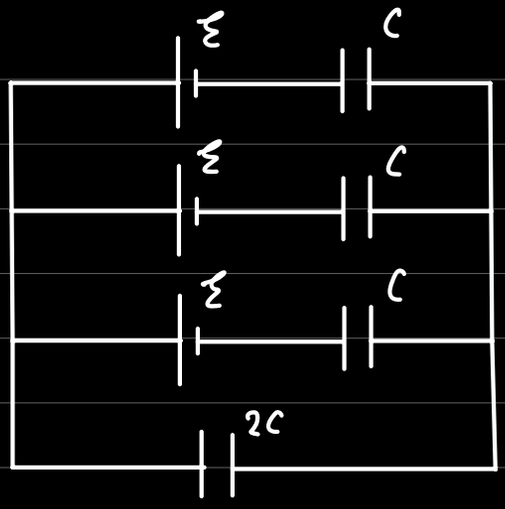
$$\mathcal{E}_{eff} = \frac{\mathcal{E}_1 C_1 + \mathcal{E}_2 C_2 + \mathcal{E}_3 C_3}{C_1 + C_2 + C_3}$$

and

$$C_{eff} = C_1 + C_2 + C_3$$

YOUTUBE.COM/@THEPHYSICSSHELTER

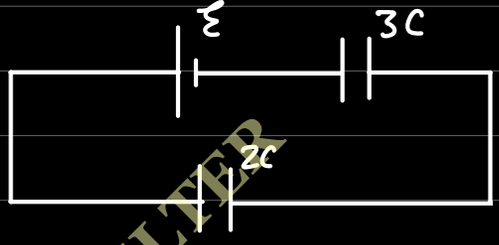
EXAMPLE: For the circuit shown in adjacent figure, find the charge on capacitor of capacitance 2C.



Solution: $\mathcal{E}_{eff} = \dfrac{\mathcal{E}C + \mathcal{E}C + \mathcal{E}C}{C+C+C} = \mathcal{E}$

$C_{eff} = C+C+C = 3C.$

This gives the simplified circuit shown in the next figure. Thus charge flown in 2C:



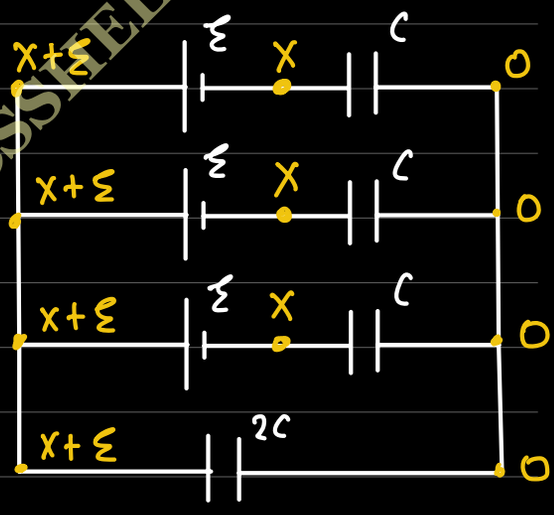
$$q = \left(\frac{2\times 3}{2+3}\right) C\mathcal{E} = \frac{6}{5} C\mathcal{E} \quad \text{Ans.}$$

Cross checking using node voltage: Applying junction law to node with 0 potential:

$$(X-0)C\times 3 + [(X+\mathcal{E})-0]\times 2C = 0$$

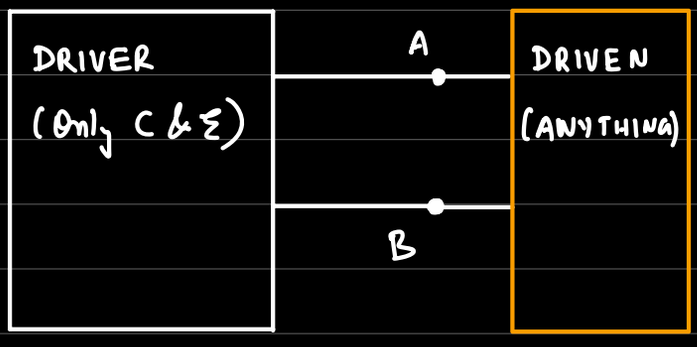
$$\Rightarrow 5XC + 2\mathcal{E}C = 0 \Rightarrow X = -2/5\,\mathcal{E}$$

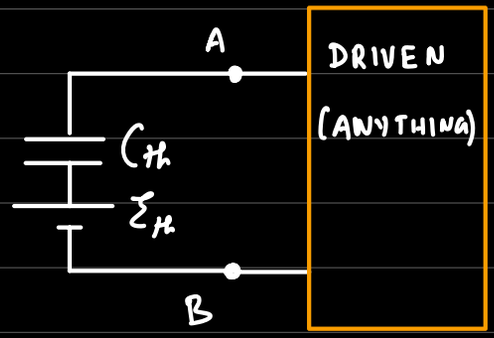
Now charge flown in 2C $= (X+\mathcal{E})\times 2C = \frac{6}{5}\mathcal{E}C.$



THEVENIN'S THEOREM:

If in any network we can divide it in to two parts such that one part (the driven) contains only capacitors and ideal batteries, while driven may contain anything and two parts are connected only through two ports (A & B) then for purpose of solving charge flows through the driven, the driven can be replaced by a single battery ($\mathcal{E}_{th}$) and a series single capacitance (C_{th}) where $\mathcal{E}_{th}$ is the open circuit voltage across terminals A and B and C_{th} is effective capacitance of the driven between opened terminals





YOUTUBE.COM/@THEPHYSICSSHELTER

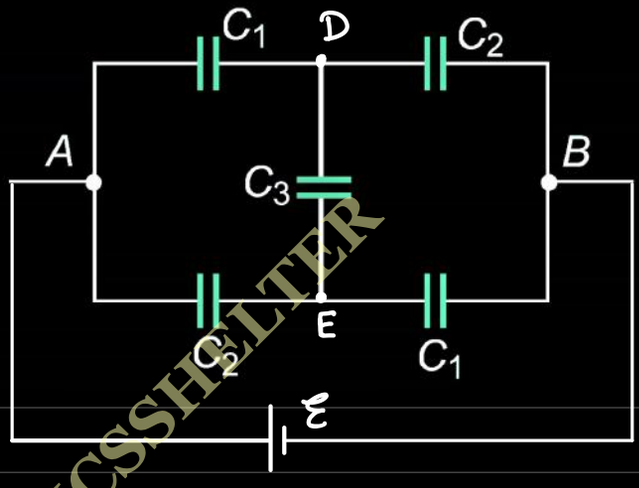
A & B (Batteries should be replaced by short circuits to find $(_H)$.
Thus:

$$\boxed{\mathcal{E}_H = V_{oc}} \qquad \boxed{C_H = C_{oc}}$$

OC: Open Circuit.

To clarify the statement and application of thevenin's theorem, we shall consider an example.

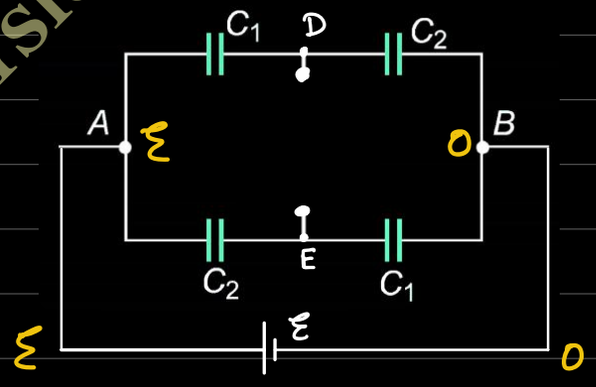
EXAMPLE: For the shown circuit, thevenize the capacitor C_3 (i.e find C_H & $\mathcal{E}_H$ across terminals D and E).



Solution: For finding $\mathcal{E}_H$ we plucked out the branch DE. Now recall that in series potential drops are inverse ratio of capacitances, thus, considering the node potentials 0 and $\mathcal{E}$, potential of junction D:

$$V_D = \frac{1/C_2}{1/C_1 + 1/C_2}\,\mathcal{E}$$

Similarly $V_E = \dfrac{1/C_1}{1/C_1 + 1/C_2}\,\mathcal{E}$



Thus $\mathcal{E}_H = V_D - V_E = \dfrac{1/C_2 - 1/C_1}{1/C_2 + 1/C_1}\,\mathcal{E} = \left(\dfrac{C_1 - C_2}{C_1 + C_2}\right)\mathcal{E}$ Ans.

Now for finding C_H, we replace battery by short circuit thus A & B nodes get shorted, making AD || DB and AE || EB. Thus between D and E, C_H can be found as series parallel simplification:

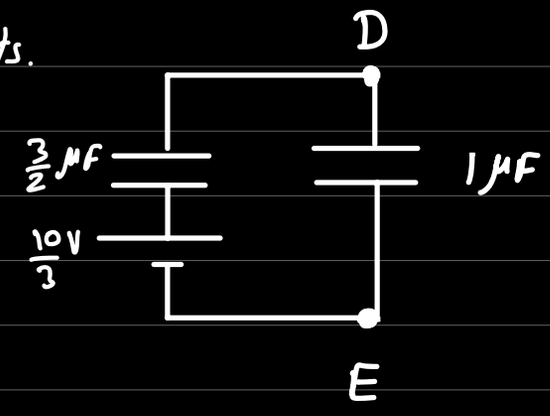
$$(C_1 \| C_2) \text{ Ser } (C_2 \| C_1) \Rightarrow C_H = \frac{C_1 + C_2}{2} \text{ Ans}$$

Think Yourself: What will be the charge on C_3 if $C_2 = C_3 = 1\mu F$ and $C_1 = 2\mu F$, and $\mathcal{E} = 10$ Volts.

YOUTUBE.COM/@THEPHYSICSSHELTER

Solution: $\mathcal{E}_{th} = ((C_1 - C_2)/(C_1 + C_2))\mathcal{E} = 10/3$ Volts.

$C_{th} = (C_1 + C_2)/2 = 3/2\ \mu F$

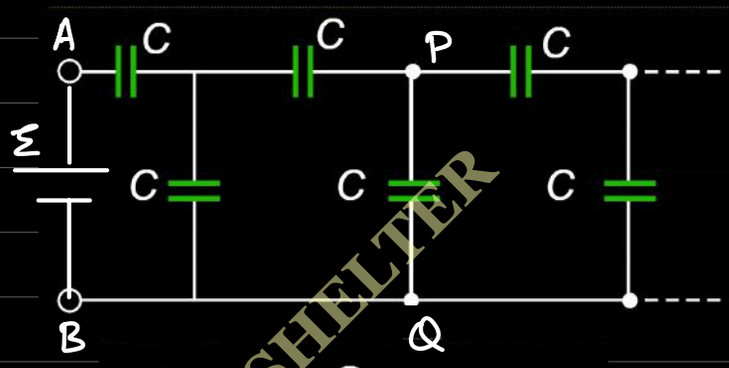


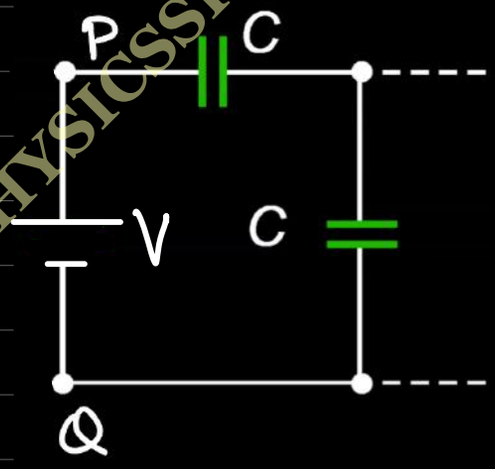
Now Charge flown $= \left(\frac{3/2 \times 1}{3/2 + 1}\right) \times \frac{10}{3}\ \mu C$

$= 2\ \mu C$ Ans.

SUBSTITUTION THEOREM:

If V is the potential drop across portion of a network connected to rest of the network through two ports (P and Q say) then the entire portion can be replaced by a single battery of V volts and it will make absolutely no difference to the charges and potentials of rest of the network. For instance in the shown infinite ladder network if V is the potential difference between P & Q then we can drop the entire network before nodes P and Q and just connect a battery of voltage V between the nodes P and Q and the network to the right of nodes P and Q will not know the difference.

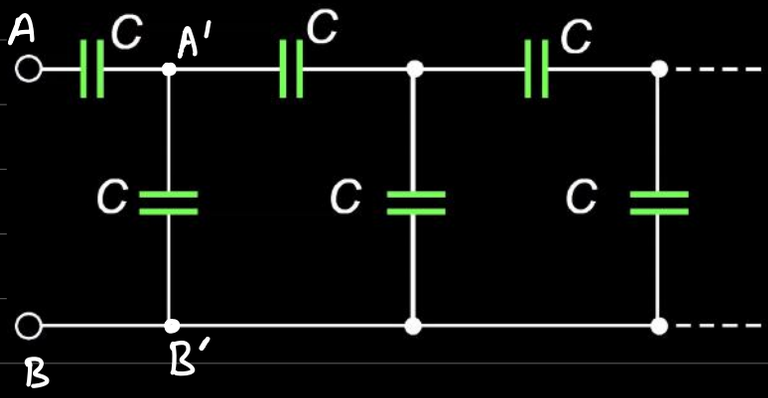




YOUTUBE.COM/@THEPHYSICSSHELTER

EXAMPLE: IRODOV 3.116: (INFINITE LADDER)

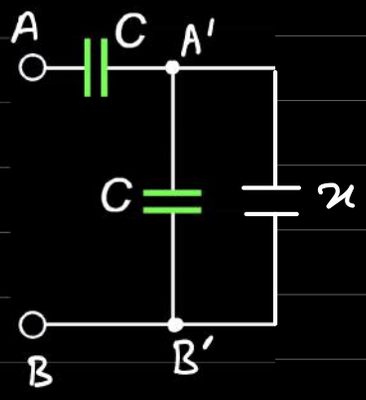
Find the capacitance of an infinite circuit formed by repetition of the same link consisting of identical capacitors each of capacitance C.



Solution: Since it is an infinite ladder, omission of finite number of units should not change the capacitance for instance if x is effective capacitance between A & B, then considering

AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

units to the right of A' and B', capacitance of infinite ladder between A' and B' must also be x. Making those replacements:



The capacitance of the new simplified circuit must also be x, thus:

$$\frac{C(C+x)}{2C+x} = x$$

$$\Rightarrow x = \left(\frac{-1 \pm \sqrt{5}}{2}\right) C, \text{ ignoring negative root,}$$

$$C_{eff} = x = \frac{\sqrt{5}-1}{2} C \text{ Ans.}$$

→ In the infinite ladder network, the potential differences and charge flows will reduce in G.P. as we move to the right. (This can be readily seen from compensation theorem).

Think Yourself: If $\mathcal{E}$ is the potential drop between A & B what will be potential drop between A' and B'?

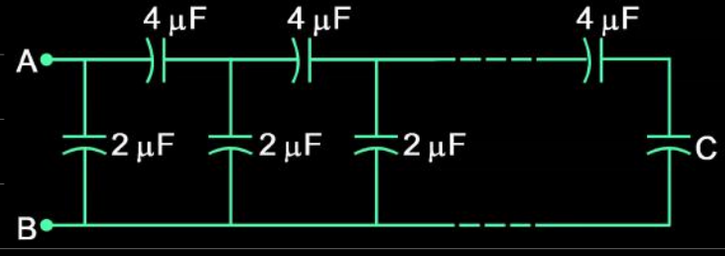
Solution: In series potential drops are in inverse ratio of capacitances

thus $$V_{A'B'}/V_{AB} = \frac{1/(C+x)}{(1/(C+x)) + (1/C)} = \left(\frac{C}{2C+x}\right) = \left(\frac{3-\sqrt{5}}{2}\right) \approx 0.38$$

Thus $$V_{A'B'} = \left(\frac{3-\sqrt{5}}{2}\right)\mathcal{E} \approx 0.38\,\mathcal{E} \text{ Ans.}$$

→ This is the common ratio of the G.P. with which charges and potential differences decline as we progress and can be used for finding P.D. across n^{th} unit.

EXAMPLE : HC VERMA: A finite ladder is constructed by connecting several sections of 2 μF and 4 μF capacitor combinations as shown in the figure. It is terminated by a capacitance C such that capacitance between A and B is independent of number of sections in between. What should be the value of C.

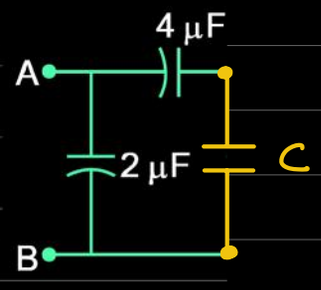


Solution: If we choose C equal to the capacitance of infinite ladder, our condition will be satisfied, because infinite number of

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

units after any finite number of units should yield same capacitance irrespective of number of prior units. Now proceeding just like previous problem:

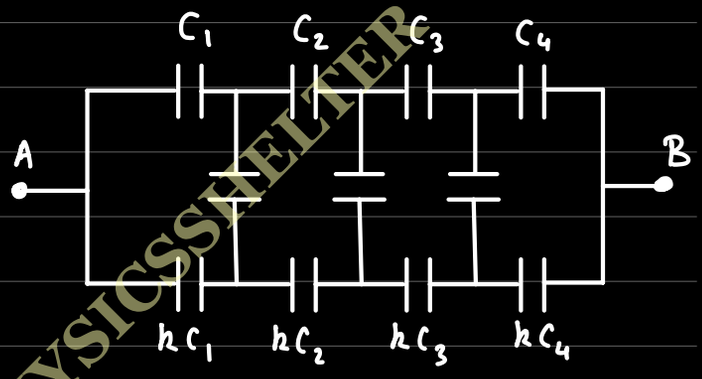
$$\frac{4C}{4+C} + 2 = C \Rightarrow C = 4\,\mu F \text{ Ans.}$$



Symmetry In Electrical Circuits:

Parallel Proportionate Branches:

If we have two parallel branches such that capacitances in one branch are proportional to that of other branch then the cross linked capacitors are redundant since they are connected between equipotential points. This is somewhat like the idea of the wheatstone bridge.



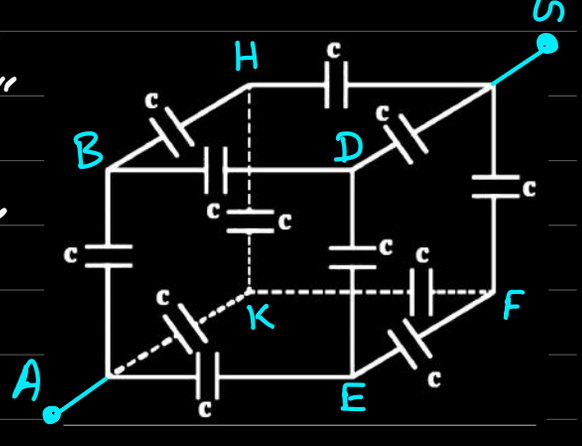
→ In any circuit if two points are equipotential then anything connected between those points (except battery) is redundant.

→ In the shown diagram, we can readily use the fact that in series, potential drops are in inverse ratio of capacitances to see that cross linked capacitors are redundant.

→ Effective capacitance for shown figure between A and B is given by: $(1/C_1 + 1/C_2 + 1/C_3 + 1/C_4)^{-1}$ [1+K]

Joining or Disconnecting Equipotential Points:

Whenever two or more points in a circuit have same potential, then joining them together makes no difference to potentials and charge flows in the circuit. For instance in the adjacent circuit, If battery is connected between A and G, B, E and K are equipotential among themselves, similarly,



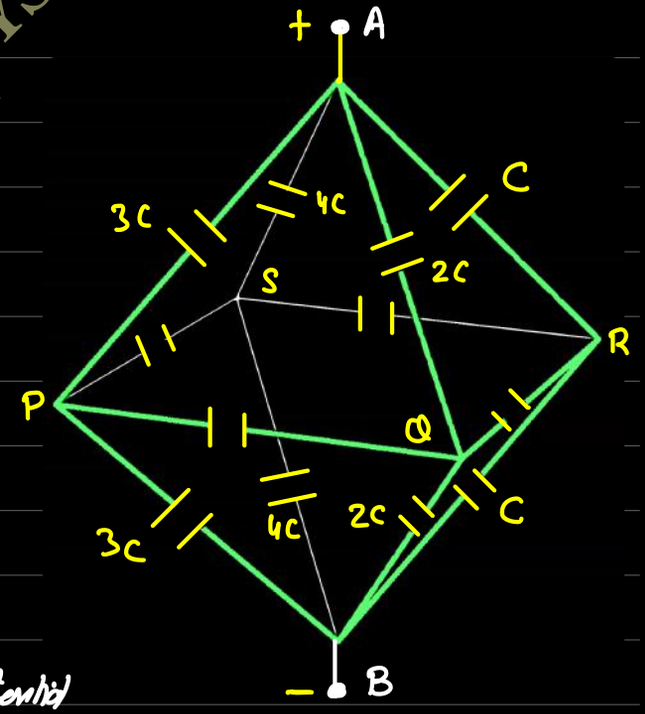
YOUTUBE.COM/@THEPHYSICSSHELTER

D, H and F are equipotential among themselves.
Thus AB, AK and AE become parallel,
similarly GD, GH and GF become parallel
and also BD, BH, ED, EF, KH and KF become
parallel to each other, this yields simplified
circuit shown in adjacent figure. Thus
Thus $C_{eff} = 6C/5$ ohms.

$$C_{eff} = \left(\frac{1}{3C} + \frac{1}{6C} + \frac{1}{3C}\right)^{-1} = \frac{6C}{5}$$

Just as joining of equipotential points is allowed similarly one can also disconnect equipotential points if they remain equipotential after disconnecting.

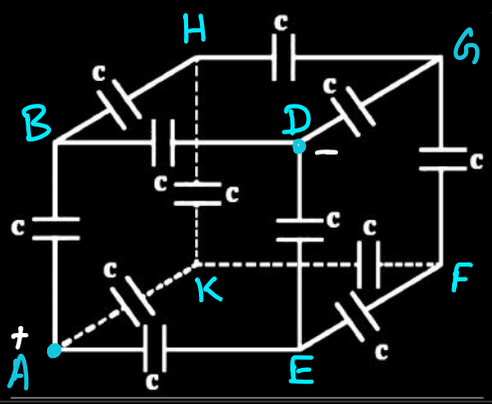
→ Redrawing of circuit can always be done to easily visualise equipotential points, thus a 3D circuit may be flattened on a 2D circuit converted to a 3D circuit, the only we need to keep the connections between the nodes intact.



SYMMETRY ABOUT EQUATOR: If the point of connection of positive terminal of battery be visualized as "North" pole and negative terminal connection point be visualised as "South Pole", then if there exists a North South Symmetry about an "equator" then potential of all points at the equator is average of potentials of North and South poles and hence all equatorial plane (on line) points are equipotential and hence their mutual connections can be ignored. (Or all points on equator can be fused together and the circuit can be visualised as series combination of "North" half and South Half.
Thus circuit in our figure is redrawn in a simplified manner as shown.
Thus $C_{eff} = 5C$ ohms.

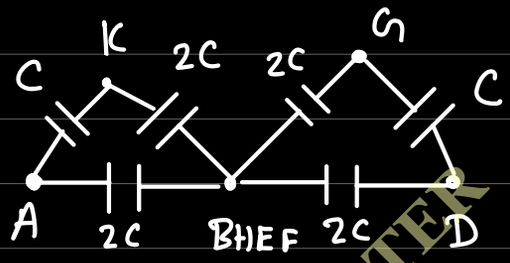
YOUTUBE.COM/@THEPHYSICSSHELTER

EXAMPLE: Find the effective capacitance of the shown cube across face diagonal AD.



Solution: If we visualise A as North pole and D as south pole then plane BHEF serves as equatorial plane. Thus BH and EF become redundant and BHEF can be represented by a single point.

Thus AB || AE, KH || KF, DB || DE and GH || GF. This yields the adjacent simplified circuit.

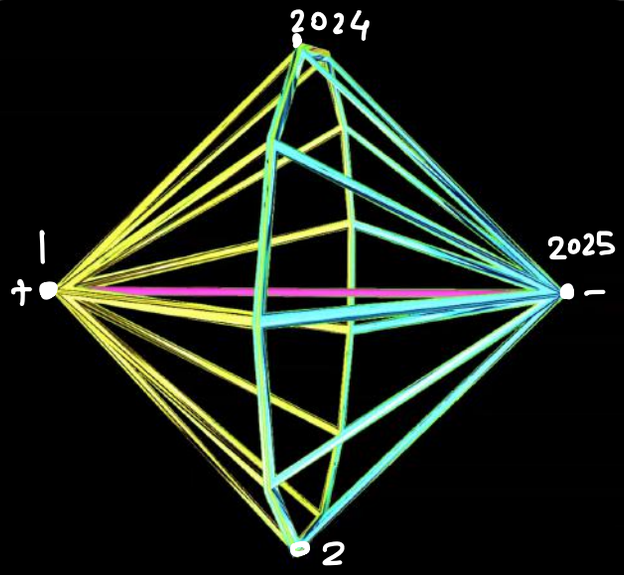


Thus $C_{eff} = \left(\frac{2}{3}C \,||\, 2C\right) \times \frac{1}{2}$

$\Rightarrow C_{eff} = 4C/3$ ohms.

EXAMPLE: There are 2025 points in space and each point is connected to each other point through a capacitor of capacitance C. Find the effective capacitance between any two points.

Solution: Since capacitance only depends on connections between the nodes and not how the nodes are laid out in space, for easy visualisation, we arrange 2023 nodes on a circle and the first and 2025th nodes as shown. Now due to equatorial symmetry, connection of 2023 points among themselves are redundant. Ignoring them, the circuit reduces to a simple series parallel arrangement.



$C_{eff} = 2023\left[\frac{C}{2}\right] + C$

[Do not forget direct connection between 1 and 2025].

Thus $C_{eff} = \frac{2025C}{2}$ ohms.

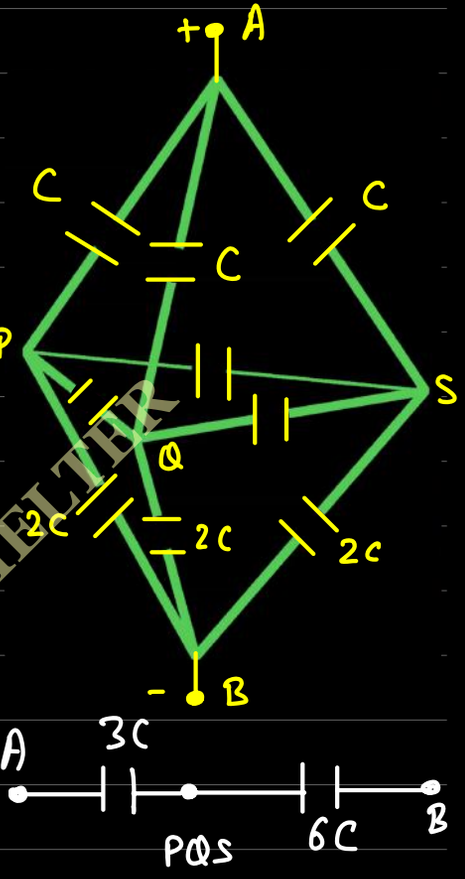
Think Yourself: Each vertex of a cube is connected to each other

YOUTUBE.COM/@THEPHYSICSSHELTER

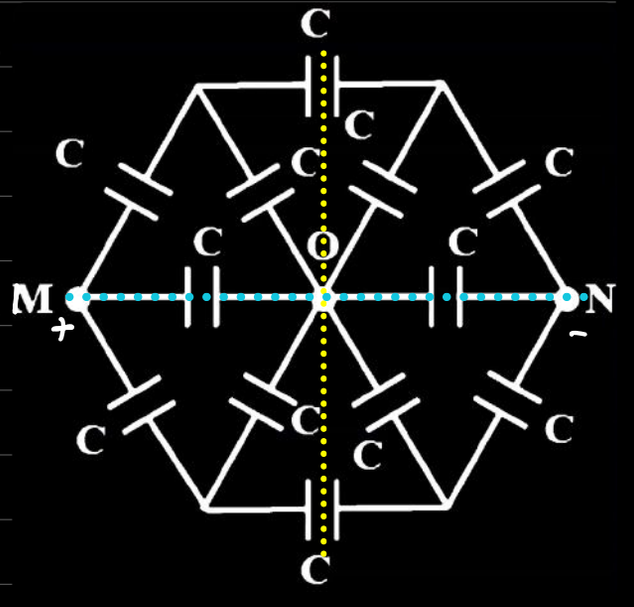
vertex through a capacitance C. Find effective capacitance across an edge of such a cube.

Solution: This is just like previous problem with $n=8$ instead of 25. Ans: $4C$.

Rotational Symmetry About Axis: Consider the adjacent figure. Here AQB and ASB branches can be thought of as being obtained by rotating the branch APB by 120° successively about axis AB. Thus we can say that there is rotational symmetry about "polar" axis AB. In such case the points on same latitude circle have same potential, thus, P, Q and S each have same potential. Thus here, the simplified circuit can be drawn as shown in the following figure. Here effective capacitance now is readily calculated as $C_{eff} = \left(\frac{3\times 6}{3+6}\right)C = 2C$ Ans.



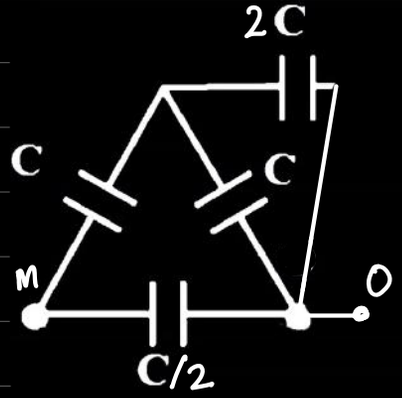
Quadrant Symmetry: Consider the circuit shown in adjacent figure. We wish to find the capacitance between M and N. This is a case of simultaneous symmetry about equator (left right symmetry, M & N one poles) and axial symmetry (up down symmetry). When this happens we can just consider the capacitance of one quadrant to get the final answer of capacitance. Here we need to carefully split the capacitors falling on quadrant boundary (halve or double based on logic of series or parallel). See the next figure.



All points on equator can be shorted with centre of symmetry after appropriately splitting the capacitors. Now the circuit is reduced to simple series and parallel combination. Thus,

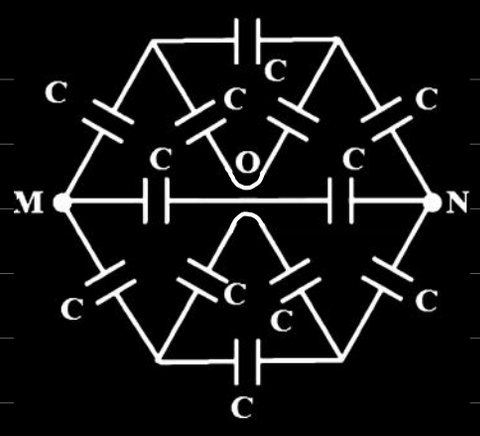
$$C_{eff} = (3C \text{ Ser } C) \,||\, (C/2) = \left(\frac{3}{4}+\frac{1}{2}\right)C$$

$$\Rightarrow C_{eff} = \frac{5}{4}C \text{ Ans.}$$



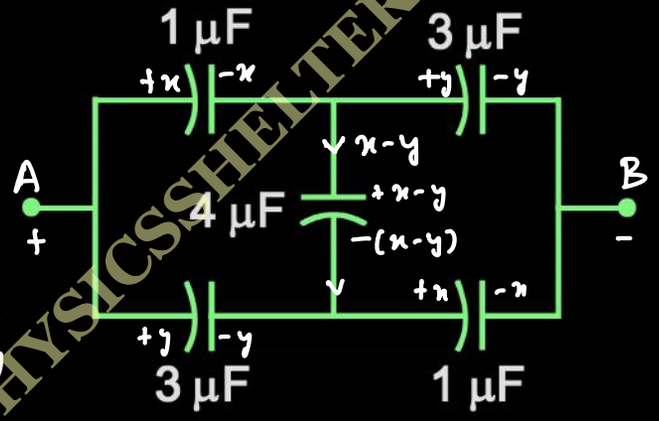
YOUTUBE.COM/@THEPHYSICSSHELTER

→ An alternate way of simplifying the circuit would be to separate the connections at the equator by displacing the connections along equator. (This works because even if the displaced points were connected, there won't be any charge flow along the connection due to points being equipotential).

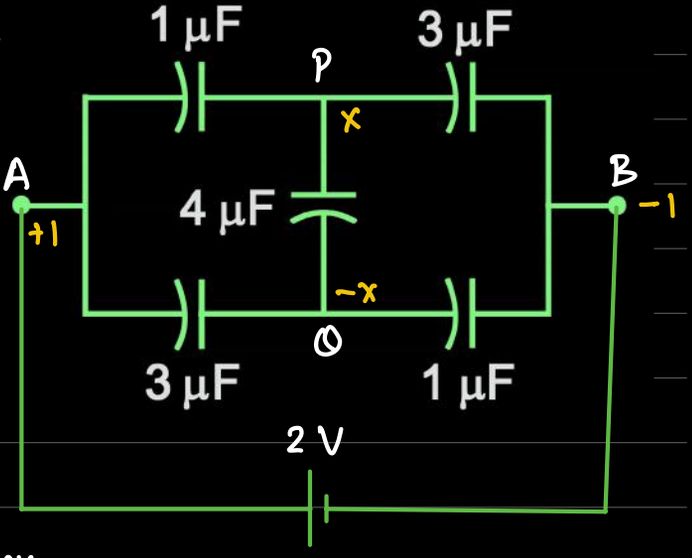


Symmetry About A Point:

Consider the circuit shown in adjacent figure. Here we can see that 180° opposite elements wrt centre of 4μF capacitor are identical. This is case of symmetry about a point. Here Top 1μF is as "close" to +Ve terminal as the lower 1μF to the −Ve terminal hence these capacitors will get symmetric (or "anti" symmetric) charges. Using this observation we can reduce the number of unknowns (Usually this problem requires three unknowns (x, y and z) recall the problem just after learning Kirchoff's laws). Similarly potential drops in symmetric branches are symmetric.



Example: For the circuit shown in the adjacent figure assume a potential of −1 Volt at node B calculate potentials of nodes P and Q, hence also find the charge flown from battery and capacitance of the system.



Solution: if we assume −1 at B, potential at A must be +1. (∵ Battery Voltage is 2V). Now we assume potential of node P as X, thus by antisymmetry, potential of Q must be −X. Now applying junction law at node P:

YOUTUBE.COM/@THEPHYSICSSHELTER

$(X-1)\times 1 + (X+1)\times 3 + [X-(-X)]\times 4 = 0$

$\Rightarrow X = -1/6$ volt, Thus $V_P = -1/6$ volt, $V_Q = +1/6$ volt.

Now charge flown from battery $q = (1-X)\times 1 + (1-(-X))\times 3 = 11/3\ \mu C$ Ans.
(∵ Considered flows along AP and AQ).

Now capacitance $= q/\varepsilon = (11/3)/2 = 11/6\ \mu F$ Ans.

CAPACITANCE OF CONCENTRIC SHELLS WITH EARTHING: Whenever we have a system of concentric shells with battery connected between any two shells and some of the shells earthed, then we assume another shell at infinity (also shorted with earth ∵ potential at ∞ is zero). Now we consider the system as a combination of concentric shell capacitors with one capacitor per concentric air gap between shells. (We will then try to draw simplified circuit by representing same potential points by a single point).

EXAMPLE: SB TIWARI : 32

Q. 32: Two concentric spherical shells have radii a and b $(>a)$. Write the capacitance of the system in following cases.

(a) Positive terminal of the battery is connected to the outer shell and its other terminal and the inner shell are grounded.

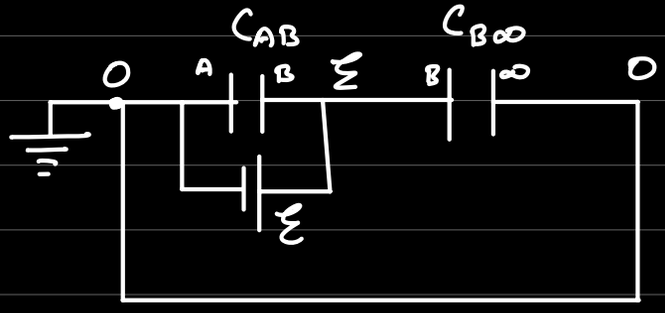
(b) Positive terminal of the battery is connected to the inner shell and its negative terminal is grounded.

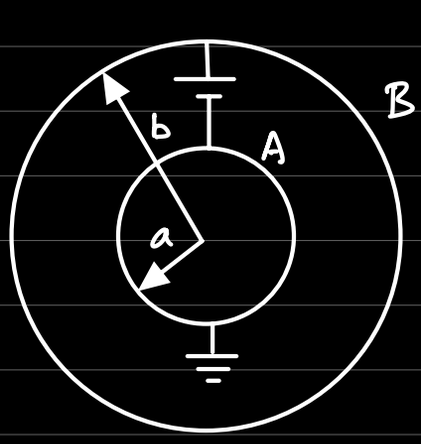
(c) A terminal of the battery is connected to the inner shell and the other terminal along with the outer shell is grounded.

(d) A terminal of the battery is connected to the outer shell and the other terminal is grounded.

YOUTUBE.COM/@THEPHYSICSSHELTER

Solution: (a) Here we visualise Two capacitors: between A and B and between B and ∞.

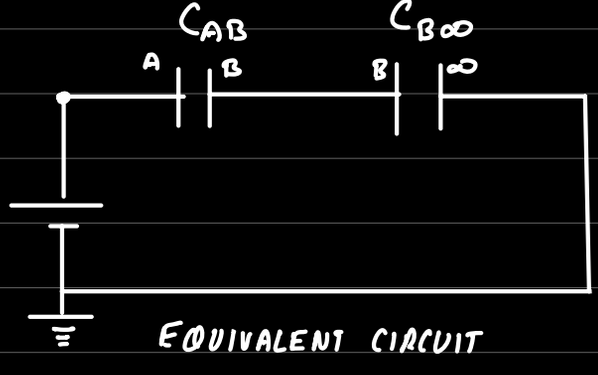




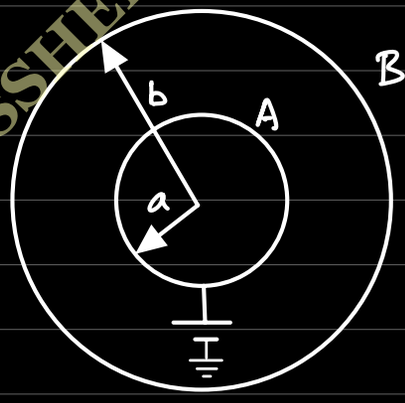
Here it can be readily seen that $C_{B\infty}$ is in parallel with C_{AB} (both are connected between O and Σ), thus $C_{eff} = C_{AB} + C_{B\infty}$

Thus $C_{eff} = 4\pi\epsilon_0 \left[\frac{ab}{b-a} + b\right] = \frac{4\pi\epsilon_0 b^2}{b-a}$ ohms

(b)



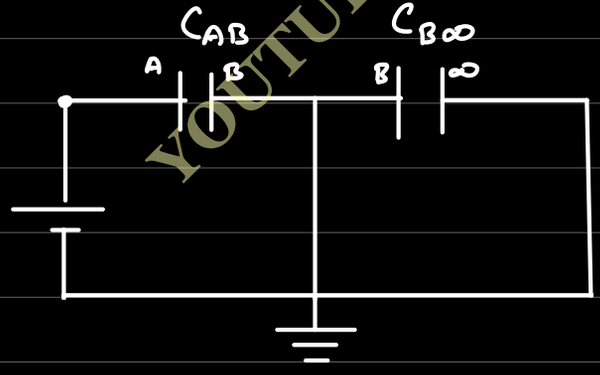
EQUIVALENT CIRCUIT

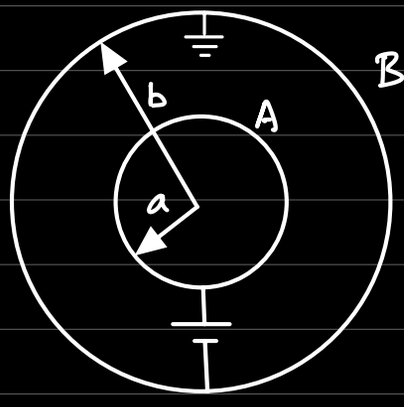


Here C_{AB} and $C_{B\infty}$ are in series as clear from the equivalent schematic diagram thus $C_{eff} = \left[\frac{b-a}{4\pi\epsilon_0 ab} + \frac{1}{4\pi\epsilon_0 b}\right]^{-1}$

$\Rightarrow$ $C_{eff} = 4\pi\epsilon_0 a$ ohms

(c)

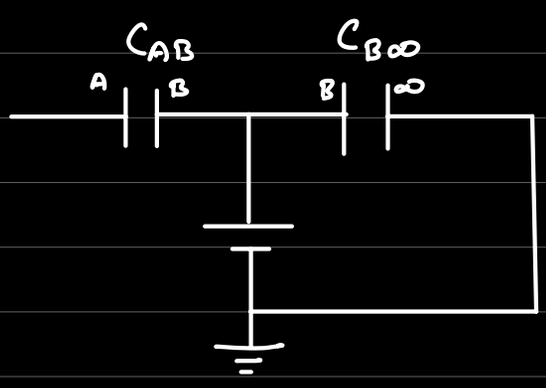


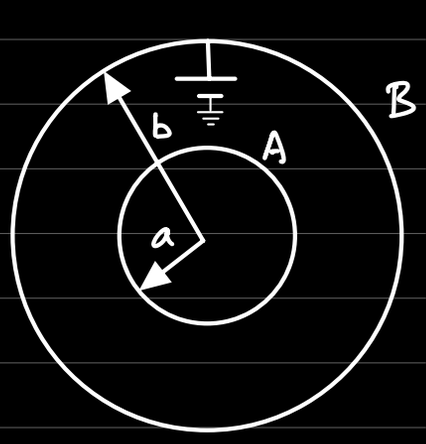


From the equivalent circuit, it is clear that the capacitor $C_{B\infty}$ is shorted thus $C_{eff} = C_{AB} = 4\pi\epsilon_0 ab/(b-a)$ ohms.

YOUTUBE.COM/@THEPHYSICSSHELTER

(d)

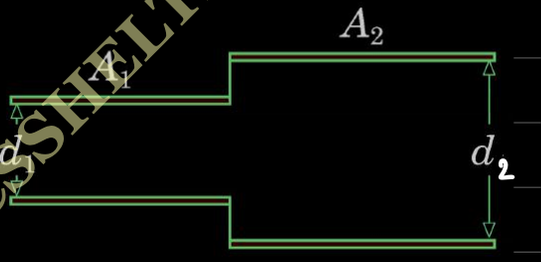




From equivalent circuit it is clear that here C_{AB} is a hanging capacitor, thus $C_{eff} = C_{B\infty} = 4\pi\epsilon_0 b$ Ans.

EXAMPLE: PATHFINDER BUILD-27

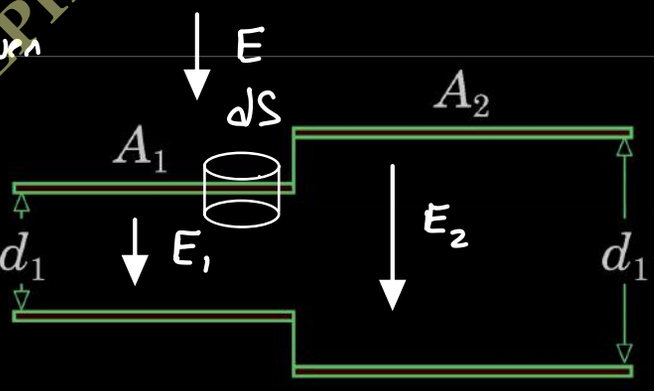
A parallel plate capacitor has two sections. In these sections, areas of plates are A_1 and A_2 and distances between the plates are d_1 and d_2 respectively. The distances between the plates are much smaller than their linear dimensions. A uniform electric field E that is perpendicular to the plates is switched on. Neglecting edge effects, deduce expressions for the intensities E_1 and E_2 of electric fields established in both the sections of the capacitor.



YOUTUBE.COM/@THEPHYSICSSHELTER

Solution: Since upper conductor and lower conductor are two equipotentials, thus:

$$E_1 d_1 = E_2 d_2 \quad (1)$$



Also $(E_1 - E)\,dS = \sigma_1\, dS/\epsilon_0$

$$\Rightarrow \quad \sigma_1 = (E_1 - E)\epsilon_0$$

Now since net combined charge of upper conductor is zero thus:

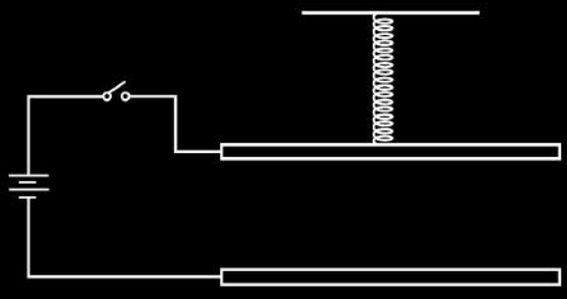
$$\sigma_1 A_1 + \sigma_2 A_2 = 0 \quad \Rightarrow \quad (E_1 - E)A_1 + (E_2 - E)A_2 = 0 \quad (2)$$

Now Solving (1) & (2): $E_1 = \dfrac{(A_1 + A_2)\, d_2 E}{A_1 d_2 + A_2 d_1}$ Ans

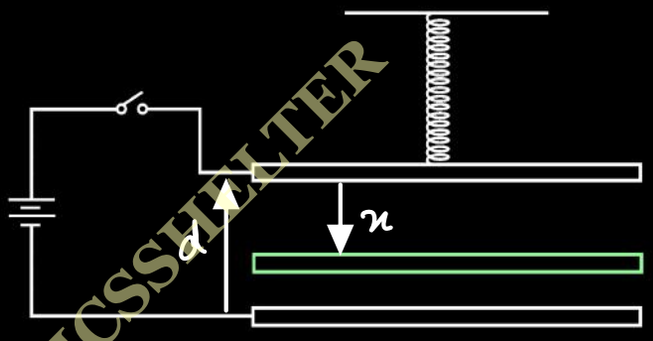
Similarly $E_2 = \dfrac{(A_1 + A_2)\, d_1 E}{A_1 d_2 + A_2 d_1}$ Ans.

Example: Pathfinder Check-17

Two identical metal plates each of area A are arranged parallel to each other as shown. The lower plate is held fixed and the upper plate is connected at its middle to one end of a spring of force constant k. The other end of the spring is attached to a fixed support. When either of the plates have no charge, the equilibrium distance between them is d. Now the switch is closed to apply a voltage between the plates. What maximum voltage in this way can be applied between the plates, so that they would not touch each other? Ignore force of gravity as compared to the spring force and the force of electrostatic interactions.



(Assume that upper plate reaches equilibrium position with negligible velocity due to some viscous damping).



Solution: For the solution we shall do the followings:

1. Find resultant force as a function of x (i.e downward displacement).
2. For equilibrium equate $F(x)=0$
3. For stable equilibrium, $\frac{dF}{dx} < 0$

(Upon increasing x, the downward force must decrease for stable equilibrium)

Now capacitance at general x, $C(x) = \frac{\epsilon_0 A}{(d-x)}$

Now $F(x) = \frac{Q^2}{2A\epsilon_0} - kx = \frac{(CV)^2}{2A\epsilon_0} - kx = \frac{AV^2\epsilon_0}{2(d-x)^2} - kx$

Now for equilibrium, $F(x) = 0$

$$\Rightarrow \frac{AV^2\epsilon_0}{2(d-x)^2} - kx = 0 \quad (1)$$

Also for stability $dF/dx < 0$ (In limiting case equality holds)

Now $$F'(x) = -k + \frac{AV^2\epsilon_0}{(d-x)^3} = 0 \quad (2)$$

Now solving (1) & (2) simultaneously: $x = d/3$ and $V = \frac{2}{3}\sqrt{\frac{2}{3}\frac{kd^3}{\epsilon_0 A}}$ Ans.

(This is upper bound of V).

YOUTUBE.COM/@THEPHYSICSSHELTER

EXAMPLE: Suppose in previous problem, applied voltage is such that equilibrium separation is d_0. (a) Find applied voltage V
(b) Find time period of oscillation of the plate.

Solution: From previous problem,

$$F(x) = \frac{\epsilon_0 A V^2}{2(d-x)^2} - kx \qquad \text{where for equilibrium } x = d - d_0.$$

Thus for equilibrium:

$$\frac{\epsilon_0 A V^2}{2 d_0^2} - k(d-d_0) = 0 \Rightarrow V = \sqrt{\frac{2k d_0^2 (d-d_0)}{\epsilon_0 A}} \quad (1)$$

Now $T = 2\pi \sqrt{\frac{m}{-F''(x)}}$ $\left(\text{Recall } T = 2\pi\sqrt{\frac{m}{u''(x)}}\right)$ and $|u''| = |F'|$

Now $F'(x) = -k + \frac{A V^2 \epsilon_0}{(d-x)^3}$ (From previous problem)

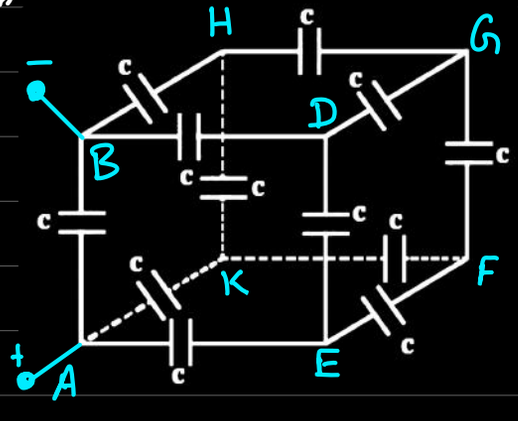
Now at $d = d_0$, $x = d - d_0 \Rightarrow F'(x) = -k + \frac{A V^2 \epsilon_0}{d_0^3}$ (2)

Now using (2) in (1):

$$F'(x) = -k + \frac{2k(d-d_0)}{d_0} = \frac{-3kd_0 + 2kd}{d_0}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{m}{-F'(x)}} = 2\pi \sqrt{\frac{m d_0}{k(3d_0 - 2d)}} \quad \text{Ans}$$

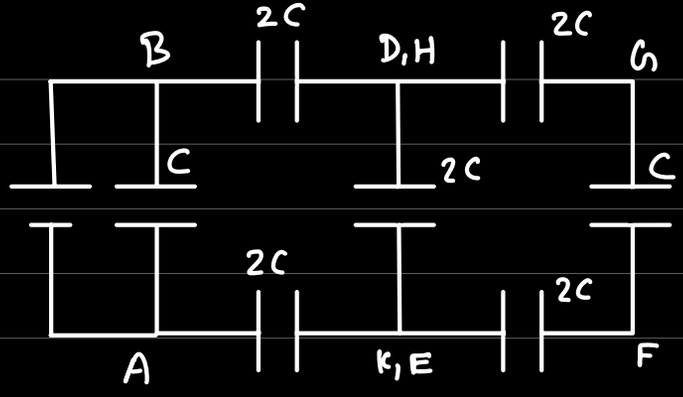
EXAMPLE: Find the effective capacitance of shown cube across the edge AB.



Solution: Recognise the case of mirror image symmetry about plane passing through AB (i.e plane ABGF). Thus D can be joined to H and E can be joined to K. Using this we can flatten the cube to draw a simplified circuit.

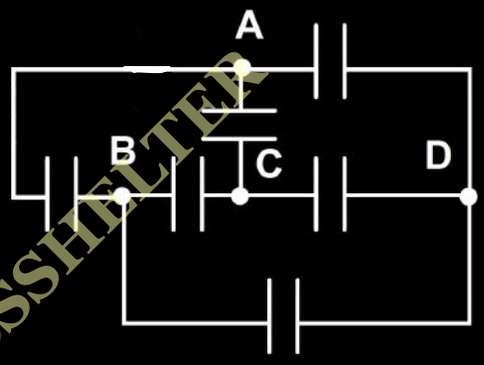
YOUTUBE.COM/@THEPHYSICSSHELTER

We make following observations:
BD ∥ BH, AK ∥ AE, HG ∥ DG, DE ∥ HC and ICF ∥ EF. This gives the adjacent equivalent circuit. This is a series parallel circuit.

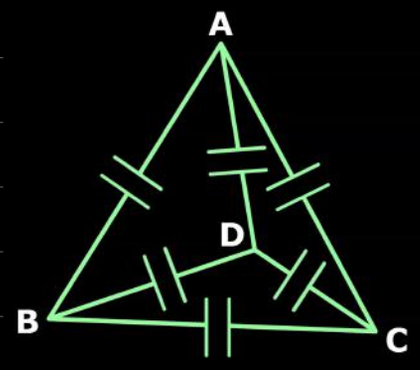


Now $C_{eff} = C \parallel (C \text{ Sen } 2{\cdot}5C) = \left[1 + \frac{2{\cdot}5 \times 1}{2{\cdot}5 + 1}\right] C = \boxed{\frac{12}{7} C}$ Ans.

EXAMPLE: KROTOV 3.13: Between which two points out of A B C and D should a battery be connected so that all capacitors (identical) get charged?



Solution: Refer to the adjacent figure which has essentially the same topology, however here in the 3D circuit it is easy to visualize that all nodes are similar and no matter where you connect the battery, one wheatstone bridge will be formed (for instance if you connect battery between B and C, branch AD will be redundant.



Think: What will be the capacitance between any two nodes if each capacitance were 1 μF.

Solution: It is a direct 1 μF parallel to a wheatstone bridge, thus effective capacitance will be 2 μF Ans.

YOUTUBE.COM/@THEPHYSICSSHELTER